When Companies Don't Die: Analyzing Zombie and Distressed Firms in a Low Interest Rate Environment*

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This draft: July 22, 2021

Abstract

We analyze the phenomenon of zombification in Europe and show that monetary policy alone is not its only driver. Concurring phenomena explain zombie and distressed firms' prevalence. Using Compustat data on public firms, we find that a rise in short-term interest rates is associated with a decrease in zombie status, suggesting that low rates constitute a favorable environment for zombie firms; there is no evidence of credit misallocation within the ECB's Corporate Sector Purchase Program; and that a decrease in inflation and a lower state of the business cycle is associated with a rise in zombie prevalence.

JEL class: D22, D24, E32, E43, E44, G33, E52

Keywords: zombie firms, distressed firms, interest rates, business cycle

*We thank Philip Valta, Marc Brunner, and participants to the 2021 Brown Bag Seminar of the Institute for Financial Management for useful comments and discussions.

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1 Introduction

With the Covid-19 pandemic, a series of government measures put into place to support European businesses might have allowed companies that would have otherwise filed for bankruptcy to continue their activities. Following the pandemic containment measures, a reduction in the number of business registrations and bankruptcies was recorded in the EU.[1] The ongoing crisis highlights the necessity to monitor and analyze the "zombification" phenomenon. That is, a constellation in which public support schemes and bank lending activities could keep non-viable firms afloat for a prolonged period of time (Laeven, Schepens, and Schnabel 2020).

The literature examines the phenomenon of zombie companies using different angles. Zombie firms are seen as the major cause of the slowdown in productivity in OECD economies (McGowan, Andrews, and Millot 2018), as a result of excessive levels of corporate debt (Jordà et al. 2020), lax monetary policies (Acharya, Eisert, et al. 2019), and erroneous bank lending behavior (Caballero, Hoshi, and Anil K. Kashyap 2008).

Insert Figure 1 here

Considering the trend in the share of zombie firms in eight European countries, Figure 1 documents an increase following crisis periods, particularly the global financial crisis. This motivates the question, are zombie companies an unintended consequence of the relaxed monetary policy put in place to mitigate recession periods? This research question, of paramount interest for policy makers, has received very little attention. This paper seeks to fill this gap. With this aim, we start by empirically analyzing the association between short-term interest rates and zombie and distressed firms' status. Next, we examine the reaction of zombie and distressed firms post announcement of the ECB's Corporate Sector Purchase Program (CSPP). The ultimate objective of monetary

[1] Eurostat on "*Fall in business registrations and bankruptcies in the EU in Q2 2020*: "`https://ec.europa.eu/eurostat/web/products-eurostat-news/-/DDN-20201104-2`.

policy measures is to support those companies that are in financial difficulties, therefore, to complement a silent literature, we expand our analysis to capture not only zombie companies, but also distressed firms. In addition, due to the interaction of monetary policy measures with the business cycle, we analyze their relationship with zombie and distress firms individually as well as in conjunction. To this end, we consider the reaction of non-viable firms following turning points in the business cycle and specific recession events. Last, we revisit the main characteristics of zombie companies and document how they differ from other categories of firms in financial difficulties, in particular the distressed firms (De Martiis, Heil, and Peter 2020).[2]

We contribute to the literature in several ways. First, using the Compustat Global Fundamentals database, a comprehensive market and corporate financial database with information regarding active and inactive publicly listed companies worldwide, we examine the link between monetary policy and non-viable firms in eight European countries. The goal of this empirical analysis is to understand, to what extent the existing low interest rate environment can affect the status of zombie and distressed firms directly or via its impact on the state of the economy. The accommodative monetary policy measures adopted by the European Central Bank in response to the crisis periods helped support productivity by easing credit conditions for non-financial companies, as well as households, and have facilitated access to credit for viable, but vulnerable firms (Obstfeld and Duval 2018). At the same time, it remains an uncovered research question whether the latter measures could increase the misallocation of capital across firms with interest rates that reached the lower bound. If interest rates are low for long, all companies should benefit, viable and non-viable, as they would have the incentive to carry on their planned investments. It has been shown that monetary easing fosters growth in more credit-constrained environments (Aghion, Farhi, and Kharroubi 2019). Nonetheless, given the existence of financial frictions, only viable firms would react to these

[2] We use the term non-viable firms to denote both zombie and distressed firms.

incentives and increase capital stock, while non-viable firms would be less responsive (Duval, Hong, and Timmer 2020). The latter scenario relates to the fact that financial frictions can generate differences in the return to capital across producers, causing losses due to misallocation (Midrigan and Xu 2014; A. V. Banerjee and Duflo 2005).

Second, since the implementation of monetary policy measures coincidences with recession periods, we augment our analysis by examining two business cycle indicators, the GDP growth and the composite leading indicator, to examine their impact on the status of non-viable firms. The composite leading indicator (CLI) anticipates turning points six to nine months before they occur and is composed of economic indicators capturing orders and inventory changes, financial market and business confidence information, and key sectors and trends data for small open economies (OECD 2021a). We complement this empirical exercise with an analysis of recession events capturing three economic downturns, the Dot-com Bubble, the Global Financial Crisis, and the Debt Crisis, to understand the distinctive effects on non-viable firms' status. Third, we use the Consumer Price Index (CPI), a measure of inflation widely observed by policy makers when setting monetary policy, to examine the effects of inflation on the zombie and distressed firms' status. Last, we complement a silent literature by investigating both zombie companies and distressed-type of firms to capture potential differences in their reactions.

We begin our empirical analysis by examining the firm-specific characteristics of zombie firms, measured following R. Banerjee and Hofmann (2020), and distressed firms identified with Altman (1968) Z-score measure. We draw upon the studies of Hoshi (2006) and Caballero, Hoshi, and Anil K. Kashyap (2008) in order to identify a firm's profitability, financial structure, size, location, and industry. We continue by investigating the relationship between the short-term interest rates, inflation, the business cycle, and the non-viable firms' status conducting logistic regressions. We then examine, whether a misallocation of credit toward non-viable firms is empirically validated, by analyzing the effect of the ECB's Corporate Sector Purchase Program (CSPP) on the non-viable firm's status using a difference-in-differences approach.

The main findings can be described as follows. In agreement with Hoshi (2006), we confirm that zombie companies are smaller in size, with lower profitability and long-term debts and liabilities. In addition, we compare the firm-specific characteristics of zombies with that of distressed firms and find that profit margin, a predictor of future profitability (Fairfield and Yohn 2001), is decisive in identifying zombie, but not distressed firms. In terms of size, both measures of the log of total assets and the log of the number of employees suggest that distressed companies are instead larger, while leverage is a predictor of both types of firms. Considering the impact of monetary policy as proxied by the short-term interest rate, our results indicate a negative and significant effect of short-term interest rates on the likelihood of being classified as zombie, supporting the argument that low interest rates could constitute a favorable environment for zombie firms livelihood (Borio 2018; R. Banerjee and Hofmann 2018). This result is not confirmed for the class of distressed firms. This can be explained by the unique features of zombie firms (De Martiis, Heil, and Peter 2020). With respect to the impact of unconventional monetary policy measures, our examination of the announcement of the CSPP and of the companies that actually participated in the program shows a significant decrease in zombie and distressed firms' status post announcement of the CSPP, consistent with Figure 1. However, at the same time, we find no evidence of misallocation of subsidies within the CSPP. These results are confirmed when using the sample of actual firms participating in the program - as published by the national central banks - as treatment group. Moreover, we find a negative and highly significant relationship between the Consumer Price Index and the non-viable firms' status, indicating that an increase in inflation can decrease the likelihood of zombie and distressed firms' status, consistent with earlier work (Bhamra, Dorion, et al. 2018; Acharya, Crosignani, et al. 2020). This finding further suggests, in line with Bhamra, Fisher, and Kuehn (2011), that the status of non-viable firms, zombie and distressed, could depend upon monetary policy through its impact on expected inflation.

Analyzing potential indirect effects of monetary policy via the business cycle, we find a negative and highly significant correlation between GDP growth and zombie and distressed firms' status, respectively. We augment this analysis with a second business cycle measure that draws upon the composite leading indicator (CLI), which forecasts turning points in economic activity by the de-trended index of industrial production (OECD 2021a). We find a positive and significant effect of the CLI on the zombie status indicating expected turning points, measured by the fluctuation of economic activity around its long-term trend, increase the zombie prevalence. Adding dummies for three recession events, the downturn following the burst of the Dot-com Bubble, the Global Financial Crisis, and the Debt Crisis, the results suggest that the probability of zombie status is significantly positively affected by recession periods, while the distressed status is not significantly affected by economic downturns. Consequently, if the number of companies undergoing zombification increases with down-turning business cycles, economies might struggle with the consequences of increasing zombie shares in addition to crisis induced economic burdens. We are, to the best of our knowledge, the first study to empirically assess the effect of the business cycle on zombie firms' status.

Taken together, the main results of this study highlight that the single monetary policy alone is likely not the main driver of zombification. Rather concurring phenomena related to the business cycle, recession events, and inflation expectation can explain the zombie firms conundrum, and, more in general, the status of non-viable firms in the eight European countries considered.

The remainder of the paper is structured as follows. Section 2 offers a brief literature review. Section 3 outlines the theoretical transmission channels of monetary policy on zombie firms. Section 4 describes the data, the construction of the non-viable firms' indicators, and provides descriptive statistics. Section 5 outlines the empirical analysis regarding the firm-specific characteristics of non-viable firms. Section 6 provides empirical evidence of the effect of monetary policy on non-viable firms' status by analyzing the impact of short-term interest rates, inflation, and the CSPP on zombie and distressed firms. Section 7 reports robustness tests and further evidence, while section 8 concludes.

2 Literature Review

This study contributes to three strands of literature. The first relates to the literature examining zombie companies, a phenomenon first investigated in reference to the Japanese banking crisis of the 1990s. Caballero, Hoshi, and Anil K. Kashyap (2008) explore zombie lending behavior, a process in which Japanese banks engaged in sham loan restructurings in order to keep the credit flowing to otherwise insolvent borrowers. As a result, an increase in zombie companies generated a depression of the investments and the employment growth of healthy companies and distortions in the creation of jobs and productivity. Peek and Rosengren (1995) provide evidence that troubled Japanese banks allocated credit to highly indebted borrowers in order to avoid the realization of losses on their balance sheets. Giannetti and Simonov (2013) exploit the Japanese banking crisis to study the effects of bank bailouts on firm access to credit, valuation, and investment and find that large capital injections can have beneficial effects on the allocation of credit. More recently, McGowan, Andrews, and Millot (2018) examine the increase in zombie companies in OECD countries and productivity performance, while Andrews and Petroulakis (2017) explore the connection between zombies and bank health confirming the weak banks channel. Looking at bank-firm relationships in Italy, Schivardi, Sette, and Tabellini (2021) confirm that zombie companies obtained credit from undercapitalized banks. Acharya, Crosignani, et al. (2020) test the zombie credit channel by examining the disinflationary effect of cheap credit toward zombie companies. In terms of identification and firm characteristics, Hoshi (2006) identifies zombie companies in Japan and compares their characteristics with those of other firms. Within these studies, we contribute by examining the firm characteristics of zombie companies and comparing them to the class of distressed firms from eight European countries.

The second strand relates to the literature investigating zombie companies and monetary policy, a less explored topic. R. Banerjee and Hofmann (2018) examine 14 advanced economies and suggest that lower interest rates tend to push up zombie shares. Borio (2018) finds persistently low interest rates allow high debt burdens to be sustainable for debtors. Examining the ECB's Outright Monetary Policy Program (OMT), Acharya, Eisert, et al. (2019) determine that post announcement of the OMT zombie companies obtained loans as banks remained undercapitalized. Gern et al. (2015) suggest that unconventional monetary policy measures may lead to a misallocation of capital as interest rates variations not only affect the level of investment, but also the structure of the investment itself. Grosse-Rueschkamp, Steffen, and Streitz (2017) examine the effectiveness of the bank-lending channel in the context of the ECB's Corporate Sector Purchase Program and confirm the role of weakly capitalized banks, especially from GIIPS countries, in the increase in lending activities to eligible firms. Igan, Hong, and Lee (2021) study how unconventional policies affect the corporate capital structure and investment decisions of Japanese firms. Altavilla, Burlon, et al. (2021) examine how sound banks pass negative rates on to their corporate depositors. We contribute to this strand by examining the relationship between monetary policy, proxied with short-term interest rates, and non-viable firms' status and their reaction post announcement of the Corporate Sector Purchase Program.

The third strand draws upon the literature examining corporate defaults and low inflation, and zombie credit and disinflation. More precisely, Bhamra, Dorion, et al. (2018) find that corporate defaults spiked in the U.S. during times of low inflation, while Acharya, Crosignani, et al. (2020) note that markets that saw an increase in zombie shares have lower inflation growth. Bhamra, Fisher, and Kuehn (2011) further suggest that corporate defaults could depend on monetary policy via its impact on expected inflation. We contribute to this literature by demonstrating that an increase in inflation can decrease the likelihood of the non-viable firms' status and suggesting that corporate defaults can depend on monetary policy via its impact on inflation. Finally, we examine whether the business cycle and specific recession-type events can further explain non-viable firms' prevalence in Europe.

3 Transmission Mechanism from Economic Theory

There are different monetary policy measures that central banks can adopt during economic downturns: conventional and unconventional monetary policy. The first operates by maneuvering nominal short-term interest rates, while the second can involve the use of negative interest rates, inflation targets and massive expansions of the central banks' balance sheets, and can attempt to influence interest rates beyond the usual short-term official rates (Joyce et al. 2012). Unconventional monetary policy programs, such as the CSPP, operate through different channels.

The default risk channel is one of the channels identified by the literature and it relates to lower grade bonds (e.g., Baa bonds) carrying a higher default risk than Treasury bonds. If the unconventional monetary policy manages to revive the economy, we can expect the default risk of corporations and investors' risk aversion to fall as the economy recovers implying a lower default risk premium (Krishnamurthy and Vissing-Jorgensen 2011). This channel is particularly relevant for the setting of this study. As the CSPP stimulates the economy by lowering borrowing costs for corporations, we would expect a reduction in non-viable firms' status.

The signaling channel, or expectation channel, rests upon the investors' expectations and affects all bond market interest rates. It is related to the concept of forward guidance, an additional monetary policy tool used to influence the expectations of market participants. Large scale asset purchases can be interpreted by market participants as a signal of poor economic times and unconventional monetary policy measures could be initiated (Bauer and Rudebusch 2013). Unconventional monetary policy can have a beneficial effect only if such policy serves as a credible commitment by the central banks to keep interest rates low (Krishnamurthy and Vissing-Jorgensen 2011).

The inflation channel, an expectation channel closely related to the signaling channel, relates to the inflation commitment of a central bank. The monthly purchases of assets are, in fact, conditional upon the central bank inflation rate target. With respect to the European Central Bank, the target was set below, but close to, 2% (European Central Bank 2016) since 2003. In July 2021, the ECB's Governing Council approved the symmetric 2% inflation target.[3] Unconventional monetary policy measures increase inflation expectations and this, in turn, can have a positive effect on nominal interest rates (Altavilla, Carboni, and Motto 2015; Krishnamurthy and Vissing-Jorgensen 2011; Krishnamurthy, Nagel, and Vissing-Jorgensen 2018).

The duration risk channel relates to the concept of the duration of a bond. In unconventional monetary policy measures, the government buys long duration assets from the private sector. This generates a reduction in the market price of duration risk thereby decreasing the yields on all long-term nominal assets including Treasuries, corporate bonds, and mortgages (Krishnamurthy and Vissing-Jorgensen 2011; Krishnamurthy, Nagel, and Vissing-Jorgensen 2018; Vayanos and Vila 2009).

The liquidity channel relates to the fact that unconventional monetary policy involves purchasing long-term securities and paying by increasing reserve balances that are more liquid than long-term securities. This generates a reduction in the price premium on the most liquid bonds and an increase in the liquidity in the hands of investors (Krishnamurthy and Vissing-Jorgensen 2011; Rodnyansky and Darmouni 2017).[4]

[3] The approval of the new ECB's strategy can be found here: `https://www.ecb.europa.eu/press/pr/date/2021/html/ecb.pr210708~dc78cc4b0d.en.html`.

[4] Other channels are at work. We refer to Krishnamurthy and Vissing-Jorgensen (2011) and Krishnamurthy, Nagel, and Vissing-Jorgensen (2018) for an excellent overall summary.

Unconventional monetary measures, such as the Quantitative Easing (QE), also affect interest rates. Three channels of transmission stand out: *1)* QE-type measures could influence the level of short-term interest rates (i.e., the overnight rate) through liquidity effects in the money market; *2)* they could contain and/or reduce the spreads that emerge in the money market and the level of the EURIBOR or LIBOR; and *3)* they could serve to manage the expectations of future monetary policy decisions and affect the slope of the money market yield curve (Lenza, Pill, and Lucrezia 2010). The expectation channel, one of the most prominent, seek to reduce the expected short-term rates and the term premium[5], respectively. This, in turn, generates a flattening of the yield curve and increases asset prices, that can trigger an improvement in the financing conditions by increasing lending, investment, consumption, and employment (Gern et al. 2015).

Unconventional monetary policy can also lead to a misallocation of capital as interest rate variations not only affect the level of investment, but also the structure of the investment itself. As a result, this could affect the future production possibilities. The longer the market interest rates are kept artificially low, the more investment decisions will be guided by distorted interest rate signals (Gern et al. 2015) and can allow high debt burdens to be sustainable for debtors (Borio 2018). When investigating large scale asset purchase programs, the credit channel also receives increasing attention as it relates to the effect of monetary policy on the flow of credit and the role of financial intermediaries - the banking counterparts. This channel encompasses the bank lending channel (Bernanke and Blinder 1988; Anil K Kashyap and Stein 1994; Peek and Rosengren 1995; Joyce et al. 2012). The latter is often put forward when explaining the phenomenon of zombie companies (Caballero, Hoshi, and Anil K. Kashyap 2008). In an environment of low interest rates, banks might be tempted to continue financing non-viable firms through the *evergreening* of loans.

[5]The term premium is known as the amount by which the yield-to-maturity of a long-term bond exceeds that of a short-term bond. The amount of such premium depends on the interest rates of the individual bonds.

4 Data and Non-Viable Firms' Characteristics

4.1 Data

Our main data source is firm-level information from Standard & Poor's (S&P) Compustat Global Annual Fundamentals file, which provides financial and market data covering about 33,000 active and inactive publicly traded companies from more than 80 countries, excluding the U.S. and Canada and includes coverage of over 96% of European market capitalization.

We matched Thomson Reuters Datastream firm ratings with Compustat data to identify a treatment and control group before the announcement of the ECB's Corporate Sector Purchase Program (CSPP) in 2016. From Datastream, we obtain information on the company name, the international securities identification number (ISIN), and the respective issuer ratings to identify companies with an investment grade rating.

We use data from the OECD databank to gather information on short-term interest rates, long-term interest rates, inflation (CPI), and the business cycle. As a business cycle measure, we use the OECD composite leading indicator (CLI) that is designed to provide early signals of turning points in business cycles showing fluctuations in economic activity around its long-term potential level (OECD, 2020).[6] As an additional measure of economic activity, we use real GDP growth data from the World Bank.

We delete observations with missing company unique identifiers (i.e., the gvkey), missing information on the fiscal year (e.g., fyear), and remove gvkey-fyear duplicates. To mitigate the effect of outliers, we winsorize our variables at the 1^{st} and 99^{th} percentile year-by-year. At the industry level, we use the GIC group based on the global industry classification standard (GICS) developed by the S&P Dow Jones Indices and the MSCI to identify 22 industries. In line with the existing literature (Acharya, Crosignani, et al. 2020; Acharya, Eisert, et al. 2019), we exclude all companies in the utilities, financial,

[6] CLI data accessed on October 9, 2020.

insurance, and banking industries. Our final sample consists of public firms from eight European countries: Italy, Spain, Greece, France, Germany, Austria, Netherlands, and Belgium over a period of 28 years, from 1990-2018. Descriptive statistics on the country sample and the industries is reported in A, while a description of the variables used in the empirical analysis is provided in Table A2.

4.2 Empirical Measures

What are zombie companies and how can we identify them? The existing literature proposes several definitions of zombie firms.[7] Kane (1989) defines them as firms whose enterprise-contributed capital has been lost. Hoshi (2006) describes them as insolvent companies with little hope of recovery, who avoid failure thanks to support from their banks. R. Banerjee and Hofmann (2018) define zombies as firms that are unable to cover debt servicing costs from current profits over an extended period. The distinctive trait lies in the financial assistance that zombie-like companies receive to remain afloat. Existing studies find that financial assistance can appear in the form of subsidized credit from banking counterparts (Caballero, Hoshi, and Anil K. Kashyap 2008), or in the form of government subsidies (Satu, Vanhala, and Verín 2020).

Consequently, the first challenge of any study investigating zombie companies lies in finding a measurement tool to classify a zombie correctly. In this respect, the literature lacks a disciplined approach. Therefore, we adopt two measures. The main measure rests on investors' expectations of a company's future prospects and classifies a company as zombie if its interest coverage ratio is below one for three consecutive years and its Tobin's q is below the median within its sector and year (R. Banerjee and Hofmann 2020).The second measure, used as robustness, follows McGowan, Andrews, and Millot (2018). A frequent mislabeling of zombie firms into other classes of companies, such as the distressed, is observable from existing studies and highlights the need of better

[7] A thorough examination is provided in A.

categorizing non-viable firms (De Martiis, Heil, and Peter 2020). In light of this and the vast literature on distressed firms, we focus our analysis on two classes of companies: zombies and distressed.

4.3 Non-Viable Firms' Status

Figure 2 documents that the zombie phenomenon appeared around the 1990s. The only exception was Italy where zombie shares are first recorded in 2000. It exhibits an increasing trend ratcheting up from the years around the global financial crisis. We confirm that the share of zombie has trended up over time and is particularly evident during economic downturns. Several peaks are registered in the eight European countries considered over a period of time that saw interest rates close to zero and the announcement of non-conventional monetary policies aimed at reviving the corporate sector.

Insert Figure 2 here

A total of 4499 firms are classified as zombie. Of these, 183 are located in Austria, 169 in Belgium, 1485 in Germany, 239 in Spain, 1200 in France, 598 in Greece, 460 in Italy, and 165 in the Netherlands. In terms of industry concentration, most of the zombie in our sample are in the industries of capital goods (631), software (538), pharmaceuticals (410), consumer durables and apparel (409), and materials (364) (Table A4).

To further understand the characteristics of the firms in our sample we monitor them over time and divide them into three categories: zombie, distressed, and healthy. The objective of this study is to examine empirically zombie firms and compare them to distressed-type of firms. The existence of different categories of non-viable firms, either zombie or distressed, underlines the difficulty ensuring that only those firms in need of financial assistance receive it. Figure 3 shows the trend of healthy (gray line), zombie (black line), and distressed firms (dashed line) from 1990 - 2018. Several conclusions can be drawn. One relates to the importance of distressed firms with respect to zombie companies, while a second aspect relates to the trend of distressed and healthy firms. In almost all of the countries the line of distressed firms suggests a downward trend, while healthy firms seem to display an upward trend. In our sample, 17170 companies are classified as healthy firms, where healthy are those that are never zombie and with a Z-score above 2.67 implying financial soundness (Altman 1968). Our sample notes 15244 companies that are classified as distressed measured with their Z-score and 611 of these are located in Austria, 607 in Belgium, 3507 in Germany, 1408 in Spain, 4690 in France, 1696 in Greece, 2057 in Italy, and 668 in the Netherlands. Figure 3 further documents the heterogeneity across countries. In Greece, from 2010 - 2015, a period of rescue programs , the share of zombie was higher than that of healthy firms, which started to drop around 2007 to recover ground only from 2016 onward.

Insert Table 1 here

In Table 1, we report the descriptive statistics for the categories of healthy, zombie, non-zombie, and distressed companies. Zombie companies fall behind their healthy peers on a number of characteristics. They show higher leverage, higher tangible assets, negative return on equity, and negative profit margins, operating profit and EBIT ICR. They are also smaller and have lower capital and research and development expenditures. The non-zombie firms can be described as a growing companies that are not as financially sound as their healthy peers. The distressed firms report higher leverage, net leverage, and higher asset tangibility with respect to the zombie firms, but at the same time they have positive returns on equity, profit margins, operating profit, and EBIT ICR.

The latter characteristics are confirmed by the vast literature on financial distress describing these companies as firms that are close to default (Gordon 1971; Altman 1968; Asquith, Gertner, and Scharfstein 1994). The two, zombie and distressed, were never compared previously. Distressed firms are also larger in size than zombie firms and have higher capital expenditures. Given these characteristics, we would expect zombie and distressed firms to react to an increase in interest rates (Section 6.1), or an increase in inflation (Section 6.2).

5 Firm-Level Analysis

In this section, we provide additional supporting evidence on the characteristics of zombie companies and, at the same time, given the large literature on distressed firms, we revisit the characteristics of zombie firms and compare them to the distressed firms in order to observe potential differences or similarities.

5.1 Non-Viable Firms' Characteristics

To analyze the characteristics of non-viable firms on a firm-level basis, we conduct a series of logistic regressions focusing on the financial structure of the company, the profitability, and its size by running the following model:

$$Pr(Status_{it} = 1) = \beta_0 + \beta_1 X'_{it} + \theta_h + \rho_c + \gamma_j + \varepsilon_{it}, \tag{1}$$

where $Status_{it}$ is the probability of company i in year t to be a zombie or a distressed company, respectively, X'_{it} is composed of firm-level explanatory variables, such as profitability, size, and leverage and these are lagged once when included in the models (Hoshi 2006). We control for industry specific effects, θ_h, country fixed effects, ρ_c, and year fixed effects, γ_j. The main dependent variable is the zombie or distressed indicator variable as defined in Section 4.2. Profitability is measured by either the profit margin that is computed as net income over sales or negative income that is an indicator variable that is equal to one whenever the return on equity is negative. Size is measured using either the log of total assets or the log of the number of employees. To capture leverage, we use two measures. The first one is computed as long-term debt plus debt in current liabilities over total assets, while the second is total liabilities over total assets.

Insert Table 2 here

Table 2 reports the results of the logistic regressions for those firms in the zombie status (Models 1 and 2) and in the distressed status (Models 3 and 4). We find that companies with lower profitability are more likely to become zombie companies as the profitability measures, profit margin, and negative income display a negative and highly significant coefficient. The results also confirm that zombie companies are more likely smaller companies as both measures of size show a negative and highly significant coefficient. In terms of financial structure, examining long-term debts and liabilities it is evident that more indebted companies have a higher probability to be identified as zombie. Both leverage measures are positive and highly significant in all models.

Differently from Hoshi (2006), we additionally compare zombies with distressed-type of firms to capture potential differences or similarities in terms of financial structure. When contrasting the results with the second set of logistic regressions using the distressed dummy as the dependent variable, we observe insignificant estimates on profit margin, while positive significant effects of negative income. Profit margin seems to be a decisive variable when comparing the determinants of zombie and distressed companies. Since profit margin is found to predict future profitability (Fairfield and Yohn 2001), one might argue that it is a specific determinant of zombie companies, who suffer from long term unprofitablility or a general lack of profitability. Our results indicate a positive effect of firm size on the probability to be classified as distressed. This constitutes a decisive difference in the clear negative impact of size on the probability to be classified as a zombie, hinting at small and clearly unprofitable companies as the stereotype of zombie firms.

6 Empirical Evidence

In this section, we investigate the effect of monetary policy on the status of non-viable firms. We start by examining whether the low interest rates can explain the behavior of non-viable firms by examining the link between the short-term interest rates and the non-viable (i.e., zombie and distressed) firms' status. Moreover, we analyze the relationship between inflation as prices by the Consumer Price Index on the prevalence of non-viable firms. Finally, we use the announcement of, as well, the actual participants in the Corporate Sector Purchase Program (CSPP) to understand the impact of this specific unconventional monetary policy measure on zombie and distressed firms' status.

6.1 Non-Viable Firms and Short-Term Interest Rates

Previous studies suggest a negative link between interest rates and zombie status, supporting the rationale that very low interest rates for long periods of time can allow high debt burdens to remain sustainable for debtors (Borio 2018; R. Banerjee and Hofmann 2018). Subsequently, non-viable firms are kept alive as lower interest rates reduce financial pressure.

Insert Table 3 here

When empirically assessing the link between interest rates and zombie or distress status, we conduct logistic regressions of the zombie and distressed status, respectively, on the short-term interest rates, controlling for profit margin, negative income, leverage, and firm size. The results are provided in Table 3. Panel A reports the results on zombie firms and reveals a negative and significant effect of interest rates on the probability of receiving classification as a zombie firm.

The level of significance decreases when including all of the controls and fixed effects, but the negative relationship remains throughout the different regressions. This result can be interpreted as supporting the argument that low interest rates potentially extend the evergreening of loans or at least constitute a favorable environment for zombie firms. However, the evidence diminishes and considering the potential links among business cycles, recession periods, and interest rates, we conducted further regressions including business cycle variables as well as recession dummies for the Dot-com Bubble, the Global Financial Crisis, and the Debt Crisis (see section 6.4) . The result confirm a negative and significant relationship between short-term interest rates and zombie status, corroborating the findings from existing studies (Borio 2018; R. Banerjee and Hofmann 2018).

With respect to the results on distressed status probability in Panel B of Table 3, we find a positive effect that turns negative and marginally significant when including the controls and fixed effects. This result is counter intuitive as higher interest rates are argued to worsen the debt burden of companies. However, when including business cycle related variables, effect becomes negligible (see section 6.4). This indicates that the state of the economy (i.e., the business cycle and and recession periods) influences significantly the distressed status rather than the level of interest rates. As the latter, as a monetary instrument, impacts the business cycle, which is shown to be a major driver of zombie and distressed status, there could well be an indirect effect. A clear identification thus remains difficult.

6.2 Non-Viable Firms and Inflation

The Consumer Price Index (CPI) is a widely used measure of inflation observed by policy makers, when setting monetary policy. In using this macroeconomic measure, we are interested in understanding how inflation influences the status of the non-viable firms in our country sample. Table 4 reports the results of a series of logistic regressions examining the relationship between CPI and the non-viable firms' status, zombie (Panel A) and distressed (Panel B), respectively.

Insert Table 4 here

From Panel A, the results document a negative and highly significant relationship in all models (1 to 3) that remains robust also in the most saturated model, model 3, which includes year, industry, country fixed effects, and firm controls. The findings suggest that an increase in inflation (CPI) decreases the probability of a firm being classified as non-viable (i.e., zombie), a relationship that is confirmed by the strand of literature examining the link between inflation and equity valuations. Bhamra, Dorion, et al. (2018) show that corporate defaults spike during times of low inflation and illustrate a strong negative relationship between expected inflation and quarterly corporate defaults in the United States between 1970 and 2016.[8] With respect to zombie-like companies, using firm-level data from a set of European countries over the period 2012-2016, Acharya, Crosignani, et al. (2020) document that markets that experience an increase in the share of zombie firms subsequently have, among other things, lower inflation growth compared to markets that have instead a lower zombie prevalence.

Considering Panel B, the results display a negative and highly significant relationship also for the distressed-type of firms in models 1 to 3, where model 3 is the most saturated regression with year, industry, country fixed effects, and firm controls.

Taken together, we can validate that corporate default decisions depend on monetary policy through its impact on expected inflation (Bhamra, Fisher, and Kuehn 2011) also in the European countries considered and, in addition, confirm that the status of non-viable firms, such as zombie companies, is affected by changes in inflation, indicating that higher inflation can lead to zombie exits.

6.3 CSPP and Non-Viable Firms' Reaction

The Corporate Sector Purchase Program (CSPP) is a non-conventional monetary policy program that forms part of the ECB's Asset Purchase Program (APP). Its objective is to ease the financing conditions in the real economy and consists of purchases of

[8] To calculate financial distress risk, the authors follow Campbell, Hilscher, and Szilagyi (2008) probability of bankruptcy.

investment-grade euro-denominated bonds implemented by the Eurosystem (De Santis et al. 2018).

Between June 2016 and December 2018, the Eurosystem conducted net purchases of corporate sector bonds under the CSPP. Outright purchases of investment-grade bonds are implemented by six Eurosystem national central banks including the Bank of Belgium, Deutsche Bundesbank, the Bank of Spain, the Bank of France, the Bank of Finland, and the Bank of Italy. Each national central bank is responsible for purchases from issuers located in a particular euro area, while the ECB coordinates the purchases. The requirements to qualify for these purchases include:[9] *1)* the issuer must be a non-financial corporation established in the euro area, defined as the location of incorporation of the issuer; *2)* the security has to have a minimum maturity of 6 months and a maximum maturity of less than 31 years; *3)* the security needs to meet the minimum requirement of a credit assessment of credit quality, BBB-/Baa3/BBBL, from at least one agency among Standard & Poor's, Moody's, Fitch Ratings or DBRS; *4)* the security must be denominated in euro and it must be issued by non-bank corporations established in the euro area. A security issued by a credit institution is not eligible for purchase; *5)* the security can be issued in the primary and secondary market; *6)* the Eurosystem applies an issue share limit of 70% per security.

The CSPP was announced in March 2016 and the purchases started in June 2016. During this period, the economy of the euro area was experiencing weak growth. We use the *announcement* of the CSPP, in a difference-in-differences (DiD) framework, as a signal to examine the reaction of the non-viable firms' status, zombie and distressed, respectively, from a country sample of eight European countries. In the vein of Grosse-Rueschkamp, Steffen, and Streitz (2017), we base our analysis on the eligibility of the investment-grade euro-denominated bonds prior to the announcement of the CSPP. Our treatment group is composed of public companies that have an investment-grade rating,

[9] Grosse-Rueschkamp, Steffen, and Streitz (2017) and De Santis et al. (2018) provide a summary of the main requirements within the Corporate Sector Purchase Program.

while the control group is composed of non-eligible public companies, (i.e., those firms that are non-investment grade rated).[10]

We conduct a DiD analysis by running the following model:

$$\begin{aligned} Pr(Status_{it} = 1) = \; & \beta_1 Post\ CSPP_t + \beta_2 Treated_i \\ & + \beta_3 Post\ CSPP \times Treated_{it} + \beta_4 X'_{it} \\ & + \theta_h + \rho_c + \gamma_j + \varepsilon_{it}, \end{aligned} \tag{2}$$

where $Status_{it}$ is the probability of company i in year t to be a zombie or a distressed, $Post\ CSPP_t$ is an indicator variable that is equal to one whenever the fiscal year is post 2016, post the announcement of the CSPP, and zero otherwise, $Treated_i$ is equal to one whenever the company is included in the treatment group as an investment-grade public company and zero otherwise, while $Post\ \ CSPP{\times}Treated_{it}$ is the difference-in-difference estimator capturing the treatment effect for company i in year t. The coefficient β_4 captures firm-specific characteristics such as size, leverage, and profitability. The latter variables enter the regression with a one year lag. To absorb country-specific, industry-specific and year-specific shocks, industry fixed effects, θ_h, country fixed effects, ρ_c, and year fixed effects, γ_j, are included in all specifications.

Insert Table 5 here

Estimation results on the logistic regressions are reported in Table 5. We observe a significant decrease in distressed and zombie status post-CSPP in Regressions 3 and 4, respectively in line with the observation from Figure 1. Whether this decrease can be due to the CSPP can be analyzed by considering the DiD estimator, $Post\ \ CSPP{\times}Treated_{it}$. Panel A reports the results using zombie status as dependent variable. For Regressions 1 - 3, we find a negative and significant effect of the DiD estimator that turns insignificant

[10]A provides additional descriptive statistics on treatment and control group.

when we include the Treatment and Post-Treatment dummies. Considering that the treatment group is based on investment-grade rated companies, including the treatment dummy is important to account for potential selection bias. Regression 4 is the most plausible set up. The insignificant DiD estimator implies that there is no evidence of misallocation of credit within the CSPP, meaning that the probability of the treatment group firms classified as zombie increases post-treatment significantly compared to the control group.

With respect to the distressed firms, Regression 4, we observe similar results, suggesting that there is no clear evidence of a reduction in the non-viable firms' status probabilities through this specific monetary policy. It is likely that the credit constrained firms are not responding to the low borrowing costs offered within the objective of CSPP. Our findings are in agreement with Obstfeld and Duval (2018) and can potentially be explained by the fact that in the presence of financial market imperfections, only those firms that are viable (i.e., not credit constrained) are likely to rapidly respond to lower borrowing costs thereby increasing their capital stock.

6.4 Non-Viable Firms' and Business Cycles

Empirical findings suggest that the rise of zombie firms is linked to economic downturns. In addition, monetary policy is commonly targeted to reduce the amplitude of the business cycles, mixing potential effects. We examine the relationship between the state of the economy and the prevalence of zombie and distressed firms by conducting logistic regressions using GDP growth as well as the CLI and recession dummies. The CLI is an indicator designed to provide early signals of turning points in business cycles showing fluctuations of the economic activity around its long-term potential level (OECD 2021a).

Insert Table 6 here

The results of the logistic regressions are displayed in Table 6. Concerning the inclusion of GDP growth, Table 6 reveals a significant negative coefficient for all models supporting the suggestion that recessions increase the probability that firms are classified as zombie, but also as distressed. In addition, the likelihood of zombie firms is positively and significantly affected by the CLI. This results is puzzling, but might be explained by the predictive property of the CLI. An increase in CLI predicts an upward trend in the business cycle, implying a currently lower state of the business cycle is related to a higher zombie share. With respect to specific recession periods, Table 6 reveals the zombie probability is significantly positively affected by all three recession periods, in the most saturated model (model 2), while the probability of being classified as distressed is not. The latter results are confirmed by further tests reported in Table A7. In all regressions, we also account for long-term interest rates as explanatory variable as the latter are considered as one of the determinants of business investment (OECD 2021b) and investment-cash flow sensitivities (Fazzari, Hubbard, and Petersen 1988; Kaplan and Zingales 1997; Cleary 1999). Precisely, low long-term interest rates encourage investment in new equipment, while high long-term rates discourage it. In turn, investments are a major source of growth.

Taken together, these findings indicate that *zombification* is a phenomenon which is aggravated by recession events. However, in contrast to distressed firms, due to the financial assistance they receive from their banking counterparts (Caballero, Hoshi, and Anil K. Kashyap 2008) or the government (Satu, Vanhala, and Verín 2020), zombies are vulnerable firms in financial difficulties for a prolonged period of time. Recessions might thus very well be the primary cause for firms to become overindebted, but they can hardly explain why these companies stay alive despite their inability to cover their debts. Consequently, it is more likely that measures and packages - whether of monetary or other nature - put in place to ease the burden of the economic crisis allow these firms to stay afloat. Considering the results of the previous sections, indicating that monetary policy measures, such as lowering interest rates, which are commonly implemented during recessions also positively impact zombie companies, it is crucial to consider the impact of all actions in conjunction. In particular in light of the COVID-19-induced economic crisis, these results emphasize the need to monitor the developments of zombie companies when considering monetary policy measures targeted at easing the economy from the burden of an endogenous shock.

Additional robustness to validate our results are reported in the Appendix, where Tables A5 and A6 test the recession events plus three years afterwards for zombie and distressed status, respectively, and confirm the baseline results from Table 6.

7 Robustness and Further Evidence

This section contains several robustness tests and provides further evidence to support the main results. We repeat the main results above presented using two alternative measures for the identification of zombie and distressed firms, respectively (Section 7.1), and an alternative CSPP treatment group of companies that actually participated in the program (Section 7.2).

7.1 Alternative Measures of Zombie and Distressed Firms

To ensure the robustness of our results, we use two alternative measures to capture zombie and distressed firms, respectively. The first measure used to classify zombie companies is based on firm accounting information and follows McGowan, Andrews, and Millot (2018). Specifically, we rely on the interest coverage ratio, a financial ratio widely used to determine the firm's ability to pay the interest on its outstanding debt, and we identify a company as zombie whenever its interest coverage ratio is below one for at least three consecutive years and its age is greater or equal to ten years.[11] This

[11] The interest coverage ratio is computed as $EBIT_{it}/IE_{it}$, where IE is interest expenses.

measure is widely used by the existing studies examining the phenomenon of zombie companies (McGowan, Andrews, and Millot 2018; R. Banerjee and Hofmann 2018).

The second measure used to identify distressed firms, widely used in the corporate finance literature to capture a firm's probability of default (Valta 2016), is based on Zmijewski (1984) model for predicting bankruptcy.[12] According to this measure, firms that have a probability greater than 0.5 are close to bankruptcy and we thus classify them as distressed. The final measures, for zombie and distressed, are two binary variables used as dependent variables in our regression models.

Insert Table 7 here

First, we reassess the link between the short-term interest rates and the zombie or distress status using both alternative measures as dependent variables for the zombie and distress status, respectively. The results in Panel A of Table 7 show a negative and highly significant relationship between the short-term interest rates and the zombie status likelihood, confirming the main findings from Table 3, and indicating that low rates for long can develop a favorable environment for the livelihood of zombie companies. Panel B of Table 7 shows that the relationship is instead negative and insignificant for the probability of the distress status, further suggesting that low interest rates are especially relevant for the zombie status while distressed firms show a negligible reaction.

[12] The measure used to identify distressed firms follows the default probability (Zmijewski 1984) calculated as:

$$N\Big(-4.336 - 4.513\Big(\frac{Net\ Income}{Total\ Assets}\Big) + 5.679\Big(\frac{Total\ Liabilities}{Total\ Assets}\Big) - 0.004\Big(\frac{Current\ Assets}{Current\ Liabilities}\Big)\Big),$$

where N is the standard cumulative normal distribution function.

Insert Table 8 here

Second, we reexamine the association between inflation and zombie or distress status using the above mentioned alternative measures for zombie and distressed firms, respectively. The results in Panel A of Table 8 document a negative and highly significant effect of the Consumer Price Index (CPI) used as a measure of inflation on the probability of the zombie status. The results largely confirm the main findings from Table 4 and, notably, confirm that the zombie status is affected by changes in inflation, signaling that higher inflation can lead to the exit of zombie companies from the market. This conclusion finds agreement in the study of Bhamra, Dorion, et al. (2018) documenting an evident spike in firms' defaults during periods of low inflation. Ultimately, in line with Bhamra, Fisher, and Kuehn (2011), we argue that corporate default decisions are determined by monetary policy via its effect on expected inflation. As before, the results are negative and insignificant for the class of the distressed firms, as reported in Panel B of Table 8. This finding can be interpreted, similarly as for the short-term interest rates, as additional evidence of the potential differences between the two classes of companies, zombie and distressed firms, respectively. Where, as visualized in Figure 1, the first are a relatively new phenomenon that spiked up following the Global Financial Crisis, while the second represent a widely covered topic in the corporate finance literature (Gordon 1971; Asquith, Gertner, and Scharfstein 1994; Campbell, Hilscher, and Szilagyi 2008).

Insert Table 9 here

Last, we retest the indirect effect of the business cycle using the alternative zombie and distressed measures as dependent variables. The results of Table 9 show that the CLI, the first measure that we use as indicator signaling turning points in the business cycle, has a negative and overall negligible effect on the zombie and distressed status likelihood, while GDP growth, a second measure used to capture economic cycle, confirms a negative effect (model 1), significant at the 5% level, on the probability of the zombie status. This result is instead not confirmed for the class of the distressed firms, again highlighting potential differences between the two classes of firms. In addition, the recession events dummies capturing three economic downturns, the Dot-com Bubble, the Global Financial Crisis, and the Debt Crisis, exhibit a positive and highly significant effect on the zombie status likelihood (models 1 and 2), confirming the relevance of the fluctuations of the economy between periods of expansion and contraction. The sign of the recession dummies coefficients is however reversed compared to the main results of Table 6. Notably, the robustness test confirms the negative effect of both short-term rates and inflation on the zombie status likelihood, confirming the main findings. Overall, these results show that the main findings of the paper are not very sensitive to the use of alternative measures of zombie and distressed firms status, respectively.

7.2 Alternative CSPP Treatment

To provide further supporting evidence of the reaction of non-viable firms following the announcement of the Corporate Sector Purchase Program, we collect data from the historical lists of securities held under the CSPP program from the ECB Asset Purchase Program database.[13] Specifically, this source provides information on the historical lists of corporate bond securities purchased under the CSPP program by the 6 Eurosystem national central banks (NCB), the Bank of Belgium, the Deutsche Bundesbank, Bank of Spain, Bank of France, Bank of Finland, and the Bank of Italy, which are responsible for the purchases from issuers located in a particular part of the euro area. The issuer, e.g., Fiat Chrysler, is a corporation established in the euro area, defined as the location of incorporation of the issuer. The historical lists of securities are available on selected dates starting from June 2017 on a weekly basis, every Friday. For each list, we have information on the NCB, the ISIN code, the issuer name, the maturity date, and the coupon rate. We download all lists from June 2017 to December 2018 and we match these

[13] This information can be found from: `https://www.ecb.europa.eu/mopo/implement/app/html/index.en.html`.

information with the sample of firms from Compustat Global Annual Fundamentals, using the ISIN code, and data on the bonds of each company from Thomson Reuters Datastream. This procedure allows us to generate a dummy variable that captures the treated companies within the CSPP program. Differently to the treatment group of Section 6.3, which is aimed at capturing the reaction following the announcement of the program, the latter information allows us to identify the firms that actually participated in the CSPP and thus to capture the effect of the monetary program on the status of non-viable firms, zombie and distressed, respectively. Within the treatment we have 85 treated companies, 20 of which are classified as zombies.

Insert Table 10 here

With this information, we retest the difference-in-difference estimation and we investigate whether this specific policy shows an effect on the status of non-viable firms. Panel A of Table 10 displays a negative and significant effect of the DiD estimator on the zombie status likelihood (models 1 to 3), that turns insignificant once we add the Post CSPP dummy, the Treated dummy, the firm controls, and the year-industry-country fixed effects. This result confirms the main findings of Table 5. Similarly, Panel B confirms a positive and significant effect on the probability of the distressed status, which as in the main results becomes negligible after adding the Post CSPP and Treated dummy, the firm controls, and year-industry-country fixed effects. Taken together, the results from both Panels, A and B, of Table 10 show a significant decrease in the zombie and distressed status, which is ultimately not confirmed by the outcome of the DiD estimator, suggesting no evidence of misallocation of credit toward non-viable firms within the specific monetary program considered. These findings further suggest, in agreement with Obstfeld and Duval (2018), that vulnerable firms are likely not reacting to the lower borrowing costs offered by the program. It is however beyond the scope of this paper to examine why.

Insert Table 11 here

Last, as additional test and further evidence, we use the DiD estimator, Post CSPP × Treated, to retest the effect of the CSPP in combination with the business cycle. Columns 1 to 3 of Table 11 show the results with respect to the zombie status likelihood and report a negative and highly significant effect of the DiD estimator (model 1) which turns insignificant once we add the firm controls and the year-industry-country fixed effects (model 2), and the explanatory variables related to the business cycle, the recession events, the short-term interest rates, the long-term interest rates, and inflation (CPI) (model 3). These results confirm the main findings and further suggest that the single monetary policy alone does not explain the zombie phenomenon. Rather, concurring factors are at play. More precisely, both business cycle indicators confirm their effect on the zombie status. The CLI, our first economic cycle measure, reconfirms a positive and significant effect on the likelihood of zombie firms. Given the anticipatory predictive property of the CLI, this result implies that a lower state of the business cycle relates to a higher zombie status likelihood, further suggesting a concurrent relationship between the two. The GDP growth, our second measure of economic cycle, reconfirms a negative and highly significant effect on the zombie status probability, suggesting that an improvement in economic outlook can translate to a decrease in zombie firms likelihood. The recession event dummies of the Dot-com Bubble and the Debt Crisis show a positive and highly significant effect on the zombie status probability, confirming the results of Table 9 and indicating that differently to the distressed firms zombie companies are especially sensitive to downward trending events. In addition, the negative and significant effect of the short-term interest rates and inflation (CPI) on the zombie status is also reconfirmed. Columns 4 to 6 of Table 11 display instead the results of the distressed status likelihood. Beside GDP growth and inflation that confirm the main findings, all other explanatory variables report a negligible result further indicating that the two classes of companies do react differently. Altogether, the tests described in this section further support the view that low interest rates for long are a breeding ground for zombie companies, but at the same time concurring business cycle activities can further explain prevalence of zombie companies in the European countries observed.

8 Conclusion

There is broad agreement within the literature that there is a large variation in zombie shares across European countries and that bank lending, especially by weakly capitalized banks, is one explanation for the allocation of capital to indebted borrowers. However, there is very little academic research as to whether the level of interest rates, the business cycle and recession periods can be potential factors explaining the prevalence of non-viable firms. Little is also known about the firm-specific characteristics of non-viable firms and how zombie companies differ from distressed firms. We focus on two categories of firms, zombie and distressed, from eight European countries to examine the relationship between business cycle and non-viable firms' status and understand whether monetary policy, a highly debated, but unclear driver, can explain the incidence of zombification.

Our findings can be summarized as follows. We confirm that zombie firms are smaller in size, have lower profitability and long-term debts and liabilities. At the same time, we determine that specific characteristics differ between zombie and distressed firms. Profit margin, a predictor of future profitability, is decisive in identifying the first class of firms, but not the second. In terms of size, distressed firms tend to be larger, while leverage predicts both types of non-viable firms. In addition, the business cycle and specific recession events, like the Global Financial Crisis, the Dot-com Bubble, and the Debt Crisis, are potential, concurring factors explaining the status of non-viable firms. Poor economic conditions increase the likelihood of classification as a zombie or distressed and turning points in economic activity, forecasted via the Composite

Leading Indicator, can signal a rise in zombie prevalence. Moreover, our empirical analysis validates the argument that lower interest rates could be a breeding ground for zombie firms, in agreement with Borio (2018) and R. Banerjee and Hofmann (2018). However, by employing the announcement of the Corporate Sector Purchase Program (CSPP), a non-conventional monetary policy program targeting the corporate sector, in a difference-in-differences framework, we find no evidence of credit misallocation to non-viable firms. We do find a significant decrease in the zombie and distressed firms' status post announcement of the CSPP, consistent with our descriptive statistics. In the context of the CSPP, these effects suggest that in the presence of financial market imperfections only viable firms are able to respond to lower borrowing costs and increase their capital stock.

Given the strong dependencies of non-viable firms on the state of the economy, our results suggest an indirect effect of lower interest rates via the business cycle. Our results provide new evidence on the existence of offsetting channels in explaining the misallocation of capital across firms or industries. A direct effect of specific policies or economic variables on the zombie share is difficult to identify as zombie companies are classified based on their persistent non-profitability. Accordingly, we encourage further studies to focus explicitly on the time series dynamics. In terms of future research, we believe the impact of the COVID-19-induced economic crisis on the zombification of European firms deserves careful inspection.

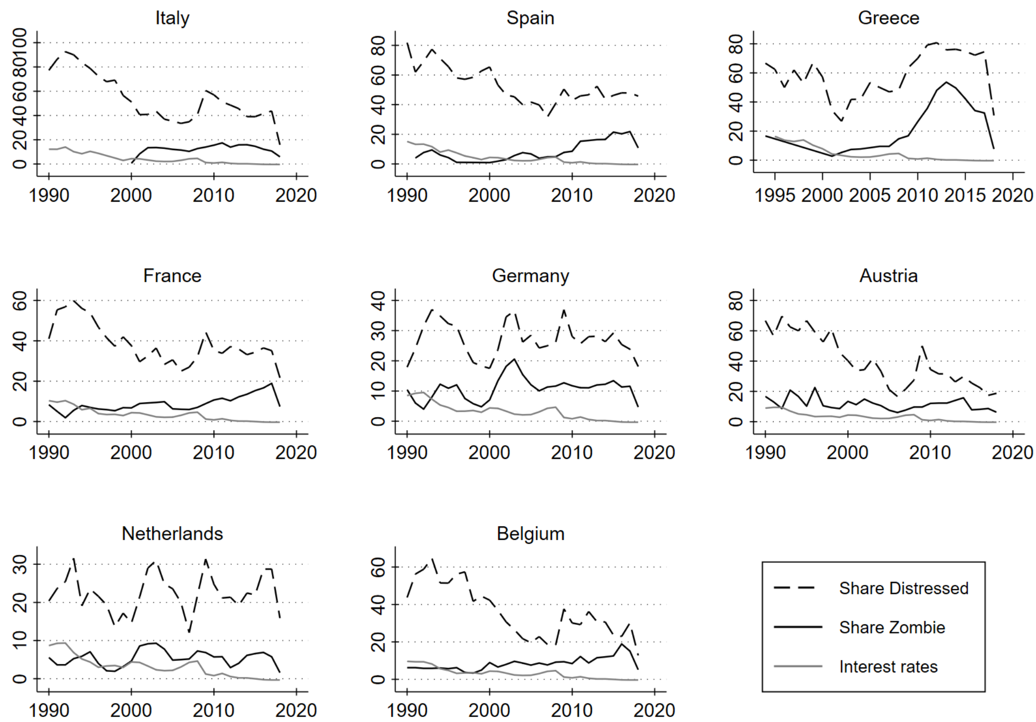

Figure 1: **Zombie Firms, Distressed Firms, and Short-Term Interest Rates.** This graph plots the shares of zombie firms, via the black line, of distressed firms, using a dashed line, and short-term interest rates, via the gray line. We define zombie firms following R. Banerjee and Hofmann (2020) and distressed firms following Altman (1968) Z-score measure. The short-term interest rates are used as a measure of the European single monetary policy, and are based on three-month money market rates, where available (OECD 2021c). Source: Authors' projections on Compustat Global data and OECD data.

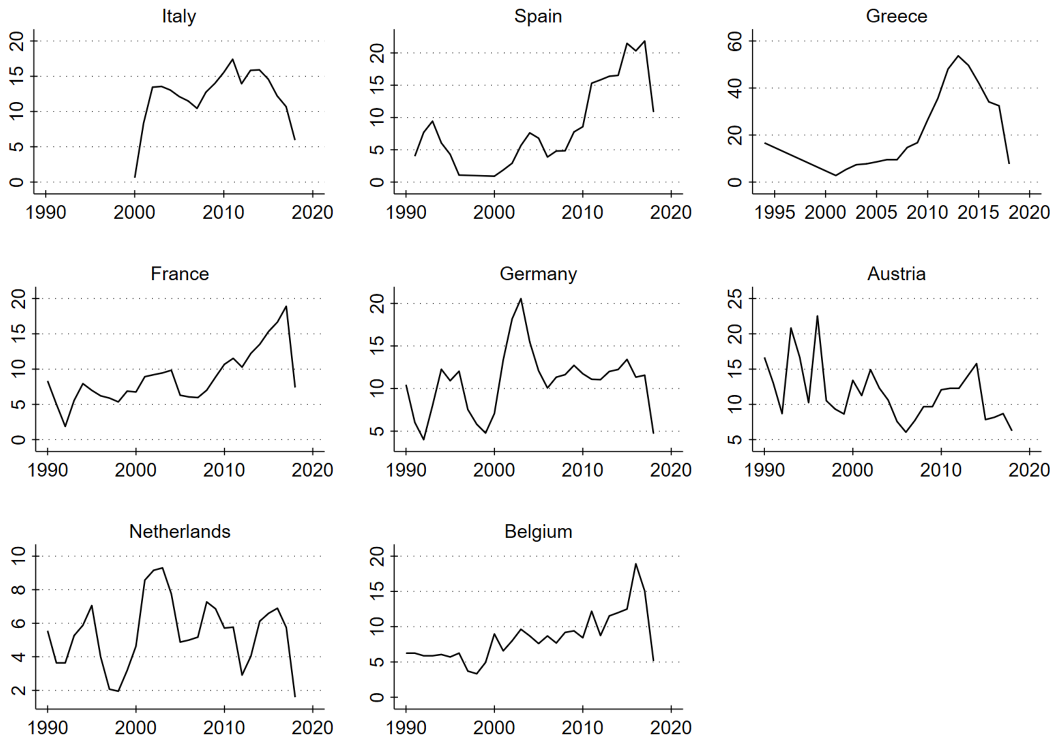

Figure 2: **Zombie Shares by Country.** This figure illustrates the mean share of zombie companies by country and year for our sample of European countries. A company is classified as zombie if its interest coverage ratio is below 1 for at least 3 consecutive years and its Tobin's q is below median (R. Banerjee and Hofmann 2020). Source: Authors' projections on Compustat data.

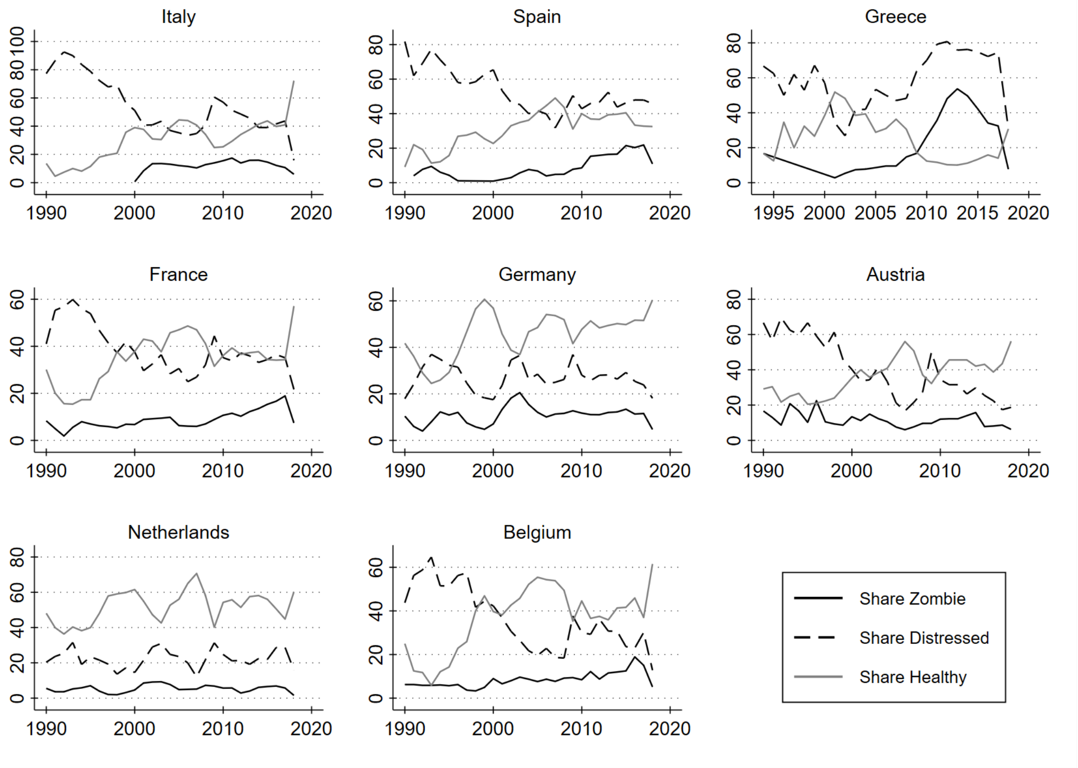

Figure 3: **Share of Zombie, Distressed, and Healthy Firms.** This figure illustrates the share of zombie, healthy, and distressed companies by country and year. A company is classified as zombie if its interest coverage ratio is below one for at least three consecutive years and its Tobin's q is below the median (R. Banerjee and Hofmann 2020). A company is classified as healthy if it has never been a zombie and has a Z-score above 2.67 (Altman 1968). Distressed firms are identified using the Z-score (Altman 1968). Source: Authors' projections on Compustat Global data.

	Healthy	**Zombie**	**Non-Zombie**	**Distressed**
Leverage	0.116	0.271	0.199	0.308
Net Leverage	-0.003	0.176	0.103	0.228
Asset Tangibility	0.139	0.175	0.190	0.246
Cash & ST Invest. Ratio	0.125	0.071	0.090	0.064
Return on Equity	0.111	-0.153	0.087	0.032
Profit Margin	0.047	-0.137	0.034	0.007
Operating Profit	0.119	-0.016	0.102	0.066
Capex Ratio	0.036	0.020	0.037	0.031
R&D Ratio	0.037	0.061	0.025	0.018
EBIT ICR	7.787	-3.529	4.320	1.534
Total Assets (Changes)	0.055	-0.070	0.041	0.012
Size Log(Tot. Assets)	4.745	4.201	5.590	6.035

Table 1: **Descriptive Statistics on Healthy, Zombie, Non-Zombie, and Distressed Firms.** This table presents the descriptive statistics for our sample of companies. We divide companies into healthy, zombie, non-zombie and distressed for descriptive purposes. Healthy companies are those that are never zombie in the entire period of observation and with a Z-score above 2.67 (Altman 1968), zombie is a binary variable that takes a value of one if its interest coverage ratio is below one for at least three consecutive years and its Tobin's q is below the median within its sector and year (R. Banerjee and Hofmann 2020). Non-zombies are those equal to zero. Distressed firms are identified using the Z-score measure (Altman 1968). Median values are reported. Source: Authors' calculations on Compustat Global data.

	Zombie		Distressed	
	(1)	(2)	(3)	(4)
Profit Margin	-0.062***	-0.041**	-0.006	0.001
	(0.019)	(0.018)	(0.008)	(0.009)
Negative Income	2.310***	2.580***	1.029***	0.917***
	(0.069)	(0.088)	(0.052)	(0.064)
Log(Tot. Assets)	-0.370***		0.129***	
	(0.024)		(0.019)	
Leverage 1	2.587***		4.007***	
	(0.201)		(0.199)	
Log(Nr. Employees)		-0.469***		0.044**
		(0.033)		(0.022)
Leverage 2		1.897***		2.949***
		(0.222)		(0.204)
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Country FE	✓	✓	✓	✓
N	25,439	16,157	25,546	16,252
R^2	0.317	0.359	0.182	0.141

Table 2: **Firm-Level Characteristics of Zombie and Distressed Firms.** This table presents the industry-country-year level logistic regressions. The dependent variables are indicator variables that are equal to one if a company is a zombie (models 1 and 2) or distressed (models 3 and 4), respectively. A company is classified as zombie if its interest coverage ratio is below one for at least three consecutive years and its Tobin's q is below the median within sector and year (R. Banerjee and Hofmann 2020). Distressed firms are classified according to (Altman 1968). Among the firm controls: profit margin is computed as net income over sales, negative income is an indicator variable that equals one whenever the return on equity is negative, size is the log of total assets, leverage 1 is the sum of long-term debt and debt in current liabilities divided by total assets, and leverage 2 is total liabilities over total assets. Standard errors in parentheses are clustered at firm-level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Panel A: Zombie	(1)	(2)	(3)
Short-Term Interest Rates	-0.396***	-1.034*	-1.007*
	(0.076)	(0.605)	(0.567)
Profit Margin		-0.091***	-0.062***
		(0.022)	(0.019)
Negative Income		2.313***	2.309***
		(0.068)	(0.070)
Leverage 1		2.560***	2.586***
		(0.203)	(0.201)
Log(Tot. Assets)		-0.353***	-0.369***
		(0.022)	(0.024)
N	41,959	25,507	25,439
R^2	0.061	0.305	0.317
Panel B: Distressed	(1)	(2)	(3)
Short-Term Interest Rates	0.059**	-0.146**	-0.143*
	(0.023)	(0.073)	(0.078)
Profit Margin		-0.002	-0.004
		(0.008)	(0.008)
Negative Income		1.026***	1.030***
		(0.051)	(0.052)
Leverage 1		3.948***	4.006***
		(0.188)	(0.198)
Log(Tot. Assets)		0.136***	0.130***
		(0.018)	(0.019)
Year FE	✓	✓	✓
Industry FE	✓		✓
Country FE	✓	✓	✓
N	41,959	25,596	25,546
R^2	0.067	0.169	0.182

Table 3: **Zombie Firms, Distressed Firms, and Short-Term Interest Rates.** This table presents logistic regressions. The dependent variable is a binary variable that is equal to 1 if a company is classified as zombie (Panel A) or as distressed (Panel B), and 0 otherwise. A company is classified as zombie following R. Banerjee and Hofmann (2020) and distressed following Altman (1968). Among the firm controls: profit margin is computed as net income over sales, negative income is an indicator variable that equals one whenever the return on equity is negative, size is the log of total assets, and leverage is the sum of long-term debt and debt in current liabilities divided by total assets. Standard errors in parentheses clustered at firm-level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Panel A: Zombie	(1)	(2)	(3)
Consumer Price Index	-0.359***	-0.182***	-0.192***
	(0.033)	(0.043)	(0.043)
Profit Margin		-0.092***	-0.063***
		(0.022)	(0.018)
Negative Income		2.304***	2.301***
		(0.068)	(0.069)
Leverage 1		2.532***	2.551***
		(0.203)	(0.201)
Log(Tot. Assets)		-0.353***	-0.370***
		(0.022)	(0.024)
N	41,965	25,507	25,439
R^2	0.065	0.306	0.318
Panel B: Distressed	(1)	(2)	(3)
Consumer Price Index	-0.087***	-0.110***	-0.112***
	(0.024)	(0.032)	(0.033)
Profit Margin		-0.003	-0.005
		(0.008)	(0.008)
Negative Income		1.017***	1.021***
		(0.052)	(0.052)
Leverage 1		3.947***	4.003***
		(0.188)	(0.199)
Log(Tot. Assets)		0.136***	0.129***
		(0.018)	(0.019)
Year FE	✓	✓	✓
Industry FE	✓		✓
Country FE	✓	✓	✓
N	41,965	25,596	25,546
R^2	0.067	0.169	0.183

Table 4: **Zombie Firms, Distressed Firms, and Inflation (CPI).** This table presents logistic regressions. The dependent variable is a binary variable that is equal to 1 if a company is classified as zombie (Panel A) or as distressed (Panel B), and 0 otherwise. A company is classified as zombie following R. Banerjee and Hofmann (2020) and distressed following Altman (1968). Among the firm controls: profit margin is computed as net income over sales, negative income is an indicator variable that equals one whenever the return on equity is negative, size is the log of total assets, and leverage is the sum of long-term debt and debt in current liabilities divided by total assets. Standard errors in parentheses clustered at firm-level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Panel A: Zombie	(1)	(2)	(3)	(4)
Post CSPP×Treated	-1.766***	-1.717***	-1.717***	0.291
	(0.448)	(0.498)	(0.498)	(0.501)
Post CSPP			-1.224***	-1.317***
			(0.232)	(0.230)
Treated				-2.085***
				(0.338)
N	41,965	25,439	25,439	25,439
R^2	0.058	0.280	0.280	0.288
Panel B: Distressed	(1)	(2)	(3)	(4)
Post CSPP×Treated	0.500***	0.458**	0.458**	-0.040
	(0.172)	(0.190)	(0.190)	(0.158)
Post CSPP			-1.107***	-0.950**
			(0.409)	(0.411)
Treated				0.528***
				(0.160)
Firm Controls		✓	✓	✓
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Country FE	✓	✓	✓	✓
N	41,965	25,546	25,546	25,546
R^2	0.067	0.174	0.174	0.177

Table 5: **CSPP Effects on Zombie Firms and Distressed Firms.** This table presents DiD regressions. The dependent variable is a binary variable that is equal to 1 if a company is classified as zombie (Panel A) or as distressed (Panel B), and 0 otherwise. A company is classified as zombie following R. Banerjee and Hofmann (2020) and distressed following Altman (1968) Z-score measure. The treatment group is composed of public companies that have an investment-grade rating, while the control group is composed of non-eligible public companies that are non-rated. Ratings information are in line with the minimum requirement of credit assessment of credit quality, BBB-/Baa3/BBBL, from at least one agency among Standard & Poor's, Moody's, Fitch Ratings or DBRS. Standard errors in parentheses are clustered at firm-level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

	Zombie		Distressed	
	(1)	(2)	(3)	(4)
Composite Leading Indicator	0.078***	0.080**	0.032**	-0.010
	(0.025)	(0.036)	(0.015)	(0.020)
GDP Growth	-0.113***	-0.096***	-0.042***	-0.043**
	(0.018)	(0.025)	(0.012)	(0.017)
Dot-com Bubble	-1.703***	5.687**	0.625***	-0.716
	(0.488)	(2.816)	(0.201)	(0.586)
Financial Crisis	-3.481***	1.472	1.321***	-0.405
	(0.774)	(1.000)	(0.306)	(0.796)
Debt Crisis	-2.847***	1.700***	1.322***	-0.345
	(0.804)	(0.600)	(0.284)	(0.813)
Short-Term Rates	-0.311***	-1.013*	0.105***	-0.085
	(0.089)	(0.613)	(0.025)	(0.074)
Long-Term Rates	0.049***	0.018	0.076***	0.025
	(0.015)	(0.018)	(0.016)	(0.018)
Consumer Price Index	-0.247***	-0.195***	-0.086***	-0.092**
	(0.037)	(0.049)	(0.025)	(0.036)
Firm Controls		✓		✓
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Country FE	✓	✓	✓	✓
N	41,833	25,438	41,833	25,545
R^2	0.072	0.321	0.070	0.184

Table 6: **Zombie Firms, Distressed Firms, Business Cycle Indicators, and Recession Events.** This table presents logistic regressions. The dependent variable is a binary variable equal to 1 if a company is a zombie (models 1 and 2) or distressed (models 3 and 4). A company is classified as zombie following R. Banerjee and Hofmann (2020) and as distressed following Altman (1968) Z-score. The Composite Leading Indicator signals turning points in the business cycle (OECD 2021a). GDP growth refers to real growth rates. Short-term rates relate to 3-month money market rates (OECD 2021c), while long-term rates refer to government bonds maturing in 10 years. Dot-com bubble (2000-2001), financial crisis (2008-2009), and debt crisis (2011-2012) are dummies for the recession events. Standard errors in parentheses are clustered at firm-level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Panel A: Zombie	(1)	(2)	(3)
Short-Term Interest Rates	-4.582***	-3.250***	-3.136***
	(0.830)	(0.186)	(0.178)
Profit Margin		-0.003	0.001
		(0.012)	(0.012)
Negative Income		1.895***	1.931***
		(0.091)	(0.094)
Leverage 1		2.622***	2.587***
		(0.237)	(0.237)
Log(Tot. Assets)		-0.123***	-0.123***
		(0.028)	(0.031)
N	34,504	24,610	24,560
R^2	0.134	0.254	0.269
Panel B: Distressed	(1)	(2)	(3)
Short-Term Interest Rates	-0.016	-0.035	-0.035
	(0.027)	(0.079)	(0.080)
Profit Margin		-0.065***	-0.085***
		(0.015)	(0.018)
Negative Income		1.200***	1.216***
		(0.048)	(0.049)
Leverage 1		4.450***	4.648***
		(0.173)	(0.182)
Log(Tot. Assets)		-0.064***	-0.054***
		(0.020)	(0.021)
Year FE	✓	✓	✓
Industry FE	✓		✓
Country FE	✓	✓	✓
N	41,959	25,596	25,546
R^2	0.206	0.174	0.189

Table 7: **Zombie Firms, Distressed Firms, and Short-Term Interest Rates.** This table presents logistic regressions. The dependent variable is equal to 1 if a company is classified as zombie or as distressed, and 0 otherwise. A company is classified as zombie following McGowan, Andrews, and Millot (2018) and distressed following Zmijewski (1984) default probability model. Among the firm controls: profit margin is computed as net income over sales, negative income is an indicator variable that equals one whenever the return on equity is negative, size is the log of total assets, and leverage is the sum of long-term debt and debt in current liabilities divided by total assets. Standard errors in parentheses clustered at firm-level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Panel A: Zombie	(1)	(2)	(3)
Consumer Price Index	-0.268***	-0.138***	-0.147***
	(0.047)	(0.051)	(0.052)
Profit Margin		-0.003	0.000
		(0.013)	(0.012)
Negative Income		1.892***	1.928***
		(0.091)	(0.094)
Leverage 1		2.598***	2.560***
		(0.236)	(0.237)
Log(Tot. Assets)		-0.122***	-0.123***
		(0.028)	(0.031)
N	34,504	24,610	24,560
R^2	0.138	0.255	0.270
Panel B: Distressed	(1)	(2)	(3)
Consumer Price Index	-0.005	-0.051	-0.045
	(0.028)	(0.033)	(0.033)
Profit Margin		-0.066***	-0.085***
		(0.015)	(0.018)
Negative Income		1.196***	1.212***
		(0.048)	(0.049)
Leverage 1		4.443***	4.641***
		(0.173)	(0.182)
Log(Tot. Assets)		-0.064***	-0.054***
		(0.019)	(0.020)
Year FE	✓	✓	✓
Industry FE	✓		✓
Country FE	✓	✓	✓
N	41,965	25,596	25,546
R^2	0.206	0.174	0.189

Table 8: **Zombie Firms, Distressed Firms, and Inflation (CPI).** This table presents logistic regressions. The dependent variable is a binary variable that is equal to 1 if a company is classified as zombie (Panel A) or as distressed (Panel B), and 0 otherwise. A company is classified as zombie following McGowan, Andrews, and Millot (2018) and distressed following Zmijewski (1984) default probability model. Among the firm controls: profit margin is computed as net income over sales, negative income is an indicator variable that equals one whenever the return on equity is negative, size is the log of total assets, and leverage is the sum of long-term debt and debt in current liabilities divided by total assets. Standard errors in parentheses clustered at firm-level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

	Zombie		Distressed	
	(1)	(2)	(3)	(4)
Composite Leading Indicator	-0.024	-0.072*	0.001	-0.000
	(0.035)	(0.043)	(0.016)	(0.024)
GDP Growth	-0.054**	-0.005	0.004	-0.018
	(0.027)	(0.031)	(0.015)	(0.019)
Dot-com Bubble	23.460***	12.258***	-2.195***	0.047
	(6.352)	(1.027)	(0.251)	(0.606)
Financial Crisis	7.729***	4.017***	-3.625***	-0.945
	(2.113)	(0.418)	(0.367)	(0.808)
Debt Crisis	5.069***	2.778***	-3.632***	-0.926
	(1.262)	(0.288)	(0.334)	(0.847)
Short-Term Rates	-5.489***	-3.081***	-0.040	-0.001
	(1.385)	(0.172)	(0.027)	(0.082)
Long-Term Rates	0.024	0.004	0.043***	0.035**
	(0.015)	(0.017)	(0.013)	(0.014)
Consumer Price Index	-0.287***	-0.171***	0.028	-0.017
	(0.056)	(0.062)	(0.028)	(0.036)
Firm Controls		✓		✓
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Country FE	✓	✓	✓	✓
N	34,504	24,560	41,833	25,545
R^2	0.140	0.270	0.204	0.190

Table 9: **Zombie Firms, Distressed Firms, Business Cycle Indicators, and Recession Events.** This table presents logistic regressions. The dependent variable is a binary variable equal to 1 if a company is a zombie (models 1 and 2) or distressed (models 3 and 4). A company is classified as zombie following McGowan, Andrews, and Millot (2018) and distressed following Zmijewski (1984) default probability model. The Composite Leading Indicator signals turning points in the business cycle (OECD 2021a). GDP growth refers to real growth rates. Short-term rates relate to 3-month money market rates (OECD 2021c), while long-term rates refer to government bonds maturing in 10 years. Dot-com bubble (2000-2001), financial crisis (2008-2009), and debt crisis (2011-2012) are dummies for the recession events. Standard errors in parentheses are clustered at firm-level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Panel A: Zombie	(1)	(2)	(3)	(4)
Post CSPP×Treated	-2.727***	-2.451**	-2.451**	-0.552
	(1.006)	(1.047)	(1.047)	(0.543)
Post CSPP			-1.257***	-1.299***
			(0.230)	(0.229)
Treated				-1.974***
				(0.655)
N	41,965	25,439	25,439	25,439
R^2	0.058	0.280	0.280	0.284
Panel B: Distressed	(1)	(2)	(3)	(4)
Post CSPP×Treated	0.698***	0.663***	0.663***	-0.013
	(0.214)	(0.216)	(0.216)	(0.170)
Post CSPP			-1.105***	-0.939**
			(0.407)	(0.422)
Treated				0.701***
				(0.186)
Firm Controls		✓	✓	✓
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Country FE	✓	✓	✓	✓
N	41,965	25,546	25,546	25,546
R^2	0.067	0.174	0.174	0.177

Table 10: **CSPP Effects on Zombie Firms and Distressed Firms.** This table presents DiD regressions. The dependent variable is a binary variable that is equal to 1 if a company is classified as zombie (Panel A) or as distressed (Panel B), and 0 otherwise. A company is classified as zombie following R. Banerjee and Hofmann (2020) and distressed following Altman (1968) Z-score measure. The treatment group is composed of those firms that have participated in the CSPP program as identified by their national central bank publication of bonds purchased within the program. Ratings information are in line with the minimum requirement of credit assessment of credit quality, BBB-/Baa3/BBBL, from at least one agency among Standard & Poor's, Moody's, Fitch Ratings or DBRS. Standard errors in parentheses are clustered at firm-level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

	Zombie			Distressed		
	(1)	(2)	(3)	(4)	(5)	(6)
Post CSPP × Treated	-2.727***	-1.323	-1.334	0.698***	0.229	0.229
	(1.006)	(1.026)	(1.040)	(0.214)	(0.205)	(0.206)
Composite Leading Indicator			0.080**			-0.009
			(0.036)			(0.020)
GDP Growth			-0.096***			-0.043**
			(0.025)			(0.017)
Dot-com Bubble			5.661**			-0.716
			(2.815)			(0.586)
Financial Crisis			1.450			-0.403
			(1.000)			(0.797)
Debt Crisis			1.679***			-0.343
			(0.600)			(0.813)
Short-Term Rates			-1.013*			-0.084
			(0.613)			(0.074)
Long-Term Rates			0.018			0.025
			(0.018)			(0.018)
Consumer Price Index			-0.196***			-0.093**
			(0.049)			(0.036)
Firm Controls		✓	✓		✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓
N	41,965	25,439	25,438	41,965	25,546	25,545
R^2	0.058	0.317	0.321	0.067	0.182	0.184

Table 11: **CSPP Effects, Zombie, Distressed Firms, Business Cycle Indicators, and Recession Events.** This table presents logistic regressions. The dependent variable is a binary variable equal to 1 if a company is a zombie (models 1 and 2) or distressed (models 3 and 4). A company is classified as zombie following R. Banerjee and Hofmann (2020) and as distressed following Altman (1968) Z-score. The variable Post CSPP × Treated is the DiD estimator composed of the interaction between the Post CSPP dummy and the Treated group of companies actually participating in the corporate program of the ECB. The Composite Leading Indicator signals turning points in the business cycle (OECD 2021a). GDP growth refers to real growth rates. Short-Term Rates relate to 3-month money market rates (OECD 2021c), while Long-Term Rates refer to government bonds maturing in 10 years. Dot-com Bubble (2000-2001), Financial Crisis (2008-2009), and Debt Crisis (2011-2012) are dummies for the recession events. The Consumer Price Index is used as a measure of inflation. Standard errors in parentheses are clustered at firm-level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

A Appendix

A.1 Descriptive Statistics

	N	Mean	Median	SD	Min	Max
Leverage 1	42232	0.231	0.205	0.195	0.000	1.659
Leverage 2	42309	0.601	0.604	0.246	0.040	2.532
Net Leverage	42231	0.088	0.108	0.293	-0.836	1.462
Asset Tangibility	42302	0.234	0.189	0.198	0.000	0.900
Cash & ST Invest. Ratio	42307	0.144	0.088	0.160	0.000	0.915
Return on Equity	28491	0.016	0.075	2.044	-12.262	150.369
Profit Margin	28257	-0.195	0.027	2.758	-39.624	168.872
Operating Profit	42163	0.076	0.093	0.147	-0.994	0.498
Capex Ratio	34234	0.050	0.035	0.053	-0.064	0.442
Ebit ICR	40543	16.594	3.437	133.935	-969.273	2202.786
Total Assets (Changes)	42259	5.761	0.032	104.988	-1.000	5597.776
Size Log(Tot. Assets)	42327	5.764	5.413	2.656	-0.358	16.278
Size Log(Nr. Employees)	29743	0.291	0.219	2.067	-5.809	5.443

Table A1: **Firm Performance Measures.** The table reports descriptive statistics of a selection of firm performance measures used to describe zombie, healthy, and distressed firms. Leverage 1 is the sum of long-term debt and debt in current liabilities divided by total assets, leverage 2 is total liabilities over total assets, net leverage is the sum of long-term debt and debt in current liabilities minus cash and short-term investments divided by total assets, asset tangibility is property, plant and equipment divided by total assets, cash & ST investment ratio is cash and short-term investments divided by total assets, roe is net income over common equity, profit margin is computed as net income over sales, operating profit is ebitda over total assets, capex ratio is capital expenditures over total assets, ebit icr is operating income after depreciation over interest expenses, total assets changes is the change in total assets, size is either the log of total assets or the log of the number of employees.

Variable	Description	Source
Zombie (R. Banerjee and Hofmann 2020)	ICR<1 for 3 yr. & Tobin's q below median within sector-year	Compustat, Datastream
Zombie (McGowan, Andrews, and Millot 2018)	ICR<1 for 3 yr. & Age $\geq$ 10 yr.	Compustat
Distressed (Altman 1968)	$1.2\times$WCap./TA+$1.4\times$RetEarn/TA +$3.3\times$EBIT/TA+$0.6\times$ME/TL+Sales/TA	Compustat
Distressed (Zmijewski 1984)	N(-4.3 -$4.5\times$NetIncome/TA+5.7 $\times$TotalLiab/TA - $0.004\times$CurrAssets/CurrLiab)	Compustat
Composite Leading Indicator	Composite of a number of indicators that provide indication of the cycle evolution	OECD
GDP Growth	Real GDP Growth (%)	World Bank
Short-Term Rates	Averages of daily rates (%) Three-month money market rates	OECD
Long-Term Rates	Averages of daily rates (%) Government bonds maturing in 10 years	OECD
Consumer Price Index	Annual growth rate (%)	OECD
Post CSPP	= 1 if fiscal year post 2016	Compustat
Treated	= 1 if firm is in treatment group as investment-grade public firm prior to the CSPP announcement	Compustat, Datastream
Treated	= 1 if firm is in treatment group as investment-grade public firm and participated in the CSPP	Compustat, Datastream ECB

Table A2: **Definition of Variables.** The table describes the variables used in the empirical analysis. Compustat is the Standard & Poor's Compustat database. Datastream is Thomson Reuters database. OECD is the Organization for Economic Cooperation and Development database. World Bank is the World Bank's database. ECB refers to the ECB database of the APP historical lists.

		N	Mean	SD	Min	Max
Austria	z	183	12.708	4.287	6.061	22.535
	d	611	44.108	15.119	16.667	69.565
	h	592	39.640	9.226	20.513	56.250
Belgium	z	169	10.200	3.678	3.333	18.919
	d	607	36.446	12.467	12.821	64.706
	h	777	44.264	9.138	5.882	61.538
Germany	z	1485	12.543	3.663	4.000	20.562
	d	3507	27.876	5.403	17.512	37.000
	h	6231	48.988	7.379	24.500	60.669
Spain	z	239	14.373	6.279	0.909	21.849
	d	1408	52.621	10.529	31.731	81.818
	h	916	35.257	7.679	9.091	49.038
France	z	1200	10.686	4.004	1.875	18.919
	d	4690	37.865	8.309	21.675	59.877
	h	4804	38.890	7.310	15.432	57.143
Greece	z	598	35.550	15.209	2.778	53.691
	d	1696	63.800	14.241	26.786	80.769
	h	704	31.315	12.393	10.067	51.852
Italy	z	460	13.557	2.361	0.610	17.413
	d	2057	51.127	14.864	15.842	92.593
	h	1523	38.198	10.395	4.545	72.277
Netherlands	z	165	6.264	2.000	1.587	9.302
	d	668	23.178	5.105	12.069	31.579
	h	1623	54.594	7.931	36.364	70.690
Total	z	4499	14.995	10.520	0.610	53.691
	d	15244	41.155	15.777	12.069	92.593
	h	17170	43.742	10.371	4.545	72.277

Table A3: **Zombie Firms, Distressed Firms, and Healthy Firms by Country.** The table presents descriptive statistics of zombie (z), distressed (d) and healthy (h) firms by country. We identify zombie firms following (R. Banerjee and Hofmann 2020), the distressed following (Altman 1968) and the healthy are those that are never zombie, distressed and have a Z-score above 2.67 as and indicator of financial soundness of the firm (Altman 1968). Source: Author's calculations on Compustat data.

	Code	Zombie	Distressed	Healthy
Energy	1010	68	391	417
Materials	1510	364	1,862	1,254
Capital Goods	2010	631	2,837	2,452
Commercial Services	2020	184	632	970
Transportation	2030	176	926	548
Automobiles & Components	2510	58	541	438
Consumer Durables & Apparel	2520	409	1,005	1,445
Hotels, Restaurants & Leisure	2530	128	609	417
Media Services	2540	125	373	377
Retailing	2550	51	97	156
Food and Staples Retailing	3010	5	9	10
Food, Beverage & Tobacco	3020	265	1,222	923
Household & Personal Products	3030	16	86	258
Healthcare	3510	214	599	735
Pharmaceuticals	3520	410	476	895
Real Estate	4040	5	98	41
Software	4510	538	1,130	2,756
Technology	4520	224	551	1,131
Semiconductors Equipment	4530	82	159	389
Telecommunication	5010	123	371	373
Entertainment	5020	277	867	911
Real Estate Investments	6010	97	127	142

Table A4: **Zombie Firms, Distressed Firms, and Healthy Firms by Industry.** The table presents descriptive statistics by industry classification. The industry code used refers to the GIC group, Compustat item *ggroup*, which is based on the global industry classification standard (GICS) and developed by the S&P Dow Jones Indices and the MSCI. Source: Author's calculations on Compustat data.

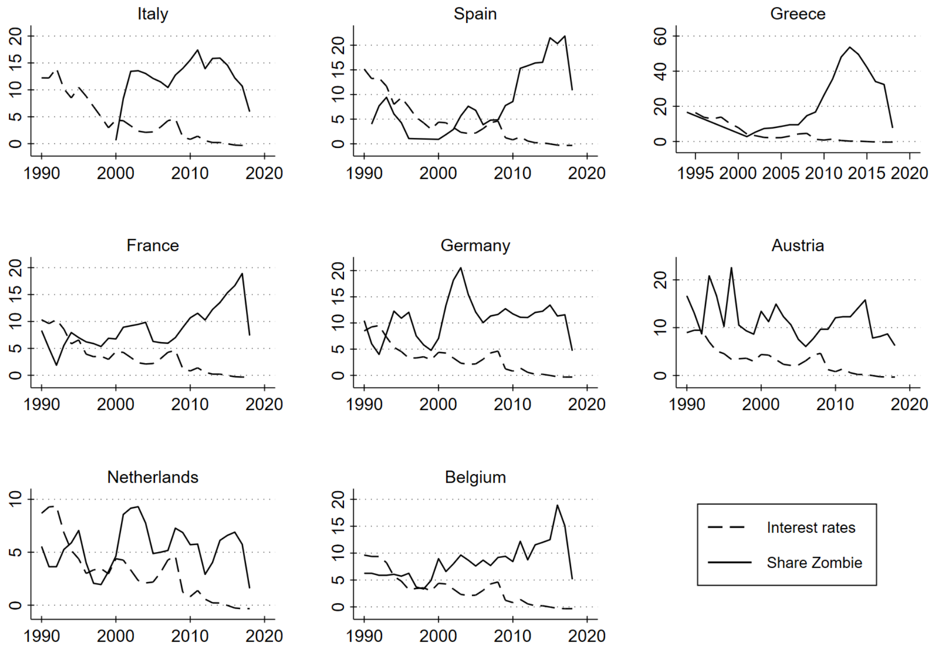

Figure A1: **Zombie Shares and Short-Term Rates.** This graph plots the share of zombie, black line, and the short-term rates, black dashed line. We identify zombie following R. Banerjee and Hofmann (2020). Source: Authors' projections on Compustat and OECD data.

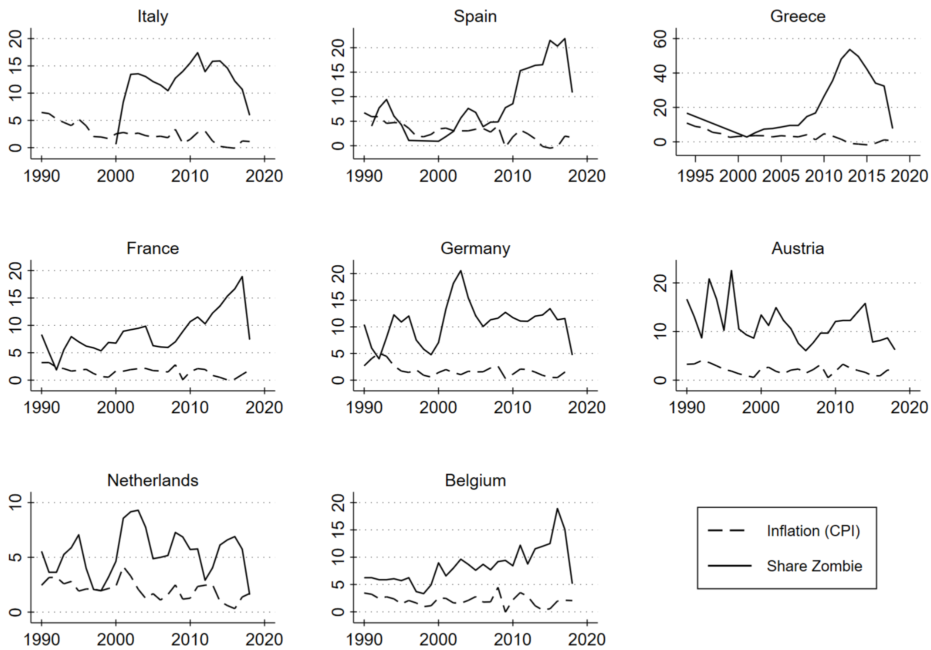

Figure A2: **Zombie Shares and Inflation.** This graph plots the share of zombie, black line, and inflation (CPI), black dashed line. We identify zombie following R. Banerjee and Hofmann (2020). Source: Authors' projections on Compustat and OECD data.

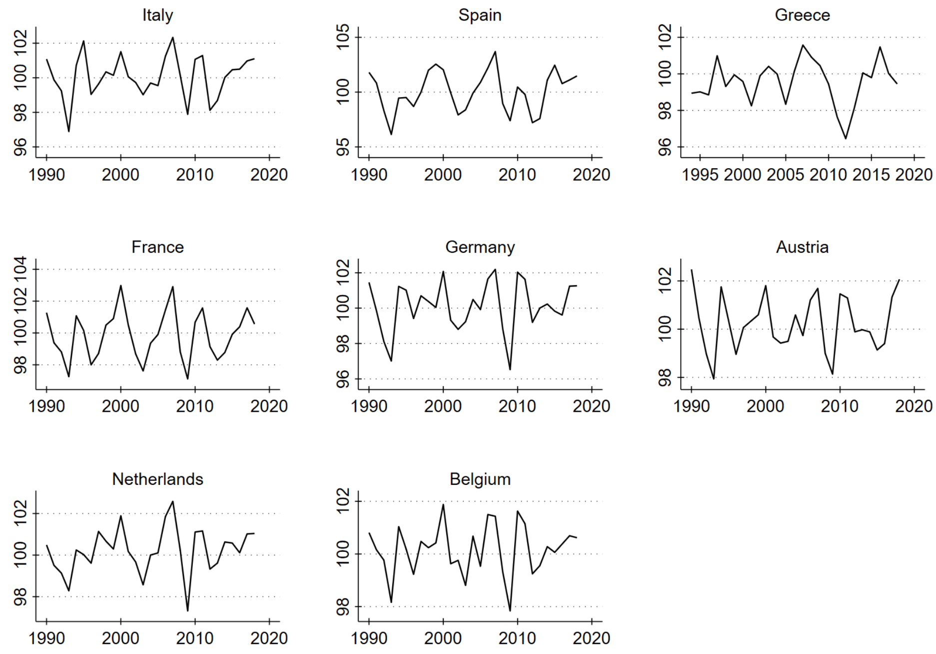

Figure A3: **Zombie Shares and Composite Leading Indicator (CLI).** This graph plots the composite leading indicator (CLI). Source: Authors' projections on OECD data.

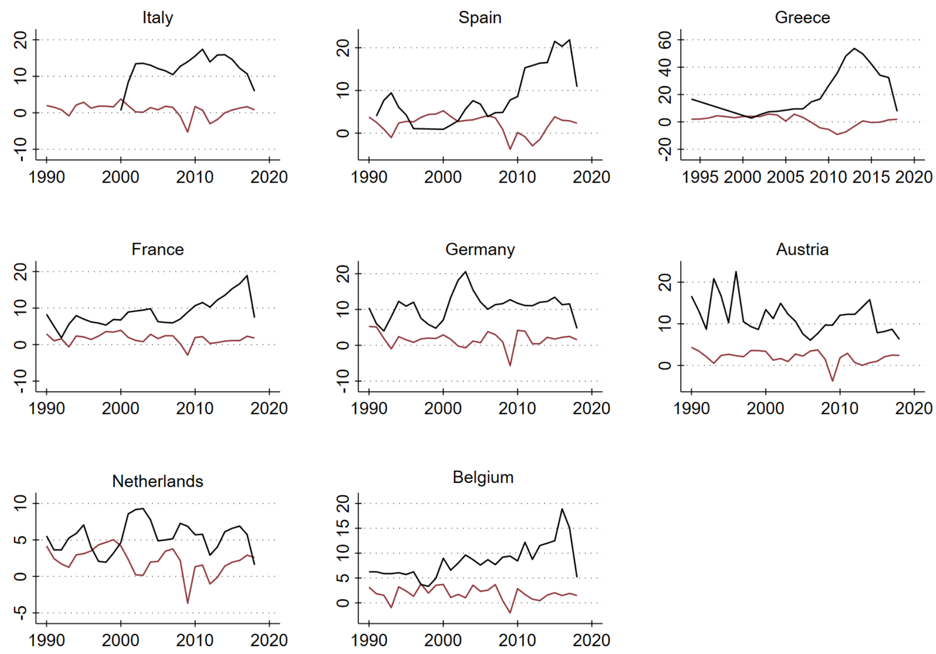

Figure A4: **Zombie Shares and GDP Growth.** This graph plots the share of zombie, black line, and GDP growth, red line. We identify zombie following R. Banerjee and Hofmann (2020). Source: Authors' projections on Compustat and World Bank data.

A.2 Additional Results

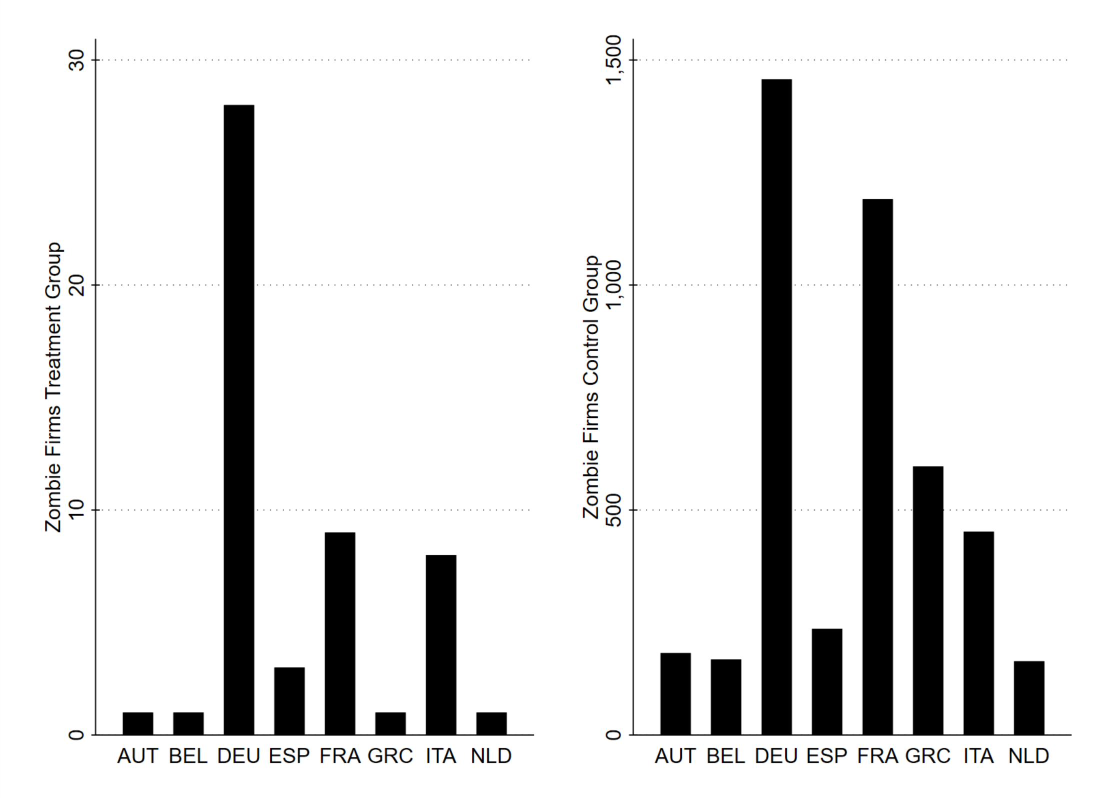

Figure A5: **(Non)-Investment Grade-Rated Zombie by Country.** The figure plots the number of investment grade-rated zombie companies by country, to the left, and the number of non-investment grade-rated zombies, to the right. We identify zombie firms following (R. Banerjee and Hofmann 2020). We use investment grade-rating information from Datastream Thomson Reuters. The majority of publicly listed companies in our country sample are non-investment grade-rated (zombie) firms. We considered the minimum requirement of a credit assessment of credit quality (BBB-/Baa3/BBBL) from at least one agency among Standard & Poor's, Moody's, Fitch Ratings or DBRS. Source: Authors' projections on Datastream and Compustat data.

	(1)	(2)	(3)
Composite Leading Indicator	0.098***	0.078***	0.080**
	(0.018)	(0.025)	(0.036)
GDP Growth	-0.102***	-0.113***	-0.096***
	(0.015)	(0.018)	(0.025)
Dot-com Bubble	0.253***	-2.211***	3.106**
	(0.059)	(0.671)	(1.506)
Financial Crisis	-0.111*	0.148	0.615
	(0.063)	(0.139)	(0.397)
Debt Crisis	0.199***	-2.995***	1.085***
	(0.048)	(0.870)	(0.313)
Short-Term Rates	-0.150***	-0.311***	-1.013*
	(0.018)	(0.089)	(0.613)
Long-Term Rates	0.087***	0.049***	0.018
	(0.012)	(0.015)	(0.018)
Consumer Price Index	-0.101***	-0.247***	-0.195***
	(0.024)	(0.037)	(0.049)
Firm Controls			✓
Year FE		✓	✓
Industry FE		✓	✓
Country FE		✓	✓
N	42394	41833	25438
R^2	0.034	0.072	0.321

Table A5: **Zombie Firms, Business Cycle Indicators, and Recession Events.** This table presents logistic regressions. The dependent variable is a binary variable that is equal to 1 if a company is a zombie and 0 otherwise. A company is classified as zombie if its interest coverage ratio is below 1 for at least 3 consecutive years and its Tobin's q is below the median within its sector and year (R. Banerjee and Hofmann 2020). The OECD Composite Leading Indicator (CLI) is an indicator signaling turning points in the business cycle. GDP growth refers to real growth rates data from the World Bank. Short-term rates relate to 3-month money market rates, while long-term rates refer to government bonds maturing in 10 years. Dot-com bubble (2000-2001), financial crisis (2008-2009), and debt crisis (2011-2012) are dummies for the recession events plus three years after. Standard errors in parentheses are clustered at firm-level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

	(1)	(2)	(3)
Composite Leading Indicator	-0.038***	0.032**	-0.010
	(0.012)	(0.015)	(0.020)
GDP Growth	-0.036***	-0.042***	-0.043**
	(0.010)	(0.012)	(0.017)
Dot-com Bubble	-0.205***	0.869***	-0.676
	(0.033)	(0.241)	(0.711)
Financial Crisis	-0.080**	0.014	0.255**
	(0.037)	(0.078)	(0.112)
Debt Crisis	0.056*	1.309***	-0.601
	(0.031)	(0.308)	(0.861)
Short-Term Rates	-0.063***	0.105***	-0.085
	(0.015)	(0.025)	(0.074)
Long-Term Rates	0.174***	0.076***	0.025
	(0.017)	(0.016)	(0.018)
Consumer Price Index	0.005	-0.086***	-0.092**
	(0.019)	(0.025)	(0.036)
Firm Controls			✓
Year FE		✓	✓
Industry FE		✓	✓
Country FE		✓	✓
N	42394	41833	25545
R^2	0.024	0.070	0.184

Table A6: **Distressed Firms, Business Cycle Indicators, and Recession Events.** This table presents logistic regressions. The dependent variable is a binary variable that is equal to 1 if a company is distressed and 0 otherwise. A company is classified as distressed if its Z-score is below 1.81 (Altman 1968). The OECD Composite Leading Indicator (CLI) is an indicator signaling turning points in the business cycle. GDP growth refers to real growth rates data from the World Bank. Short-term rates relate to 3-month money market rates, while long-term rates refer to government bonds maturing in 10 years. Dot-com bubble (2000-2001), financial crisis (2008-2009), and debt crisis (2011-2012) are dummies for the recession events plus three years after. Standard errors in parentheses are clustered at firm-level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Panel A: Zombie	(1)	(2)	(3)	(4)	(5)	(6)
Dot-com Bubble	0.102	0.544**				
	(0.186)	(0.247)				
Financial Crisis			0.336*	0.879***		
			(0.185)	(0.234)		
Debt Crisis					0.525***	1.078***
					(0.185)	(0.235)
Profit Margin		-0.062***		-0.062***		-0.062***
		(0.019)		(0.019)		(0.019)
Negative Income		2.310***		2.310***		2.310***
		(0.069)		(0.069)		(0.069)
Leverage 1		2.587***		2.587***		2.587***
		(0.201)		(0.201)		(0.201)
Log(Tot. Assets)		-0.370***		-0.370***		-0.370***
		(0.024)		(0.024)		(0.024)
N	41965	25439	41965	25439	41965	25439
R^2	0.057	0.317	0.057	0.317	0.057	0.317
Panel B: Distressed	(1)	(2)	(3)	(4)	(5)	(6)
Dot-com Bubble	-0.324***	-0.087				
	(0.111)	(0.404)				
Financial Crisis			0.083	0.501		
			(0.111)	(0.402)		
Debt Crisis					-0.063	0.313
					(0.113)	(0.402)
Profit Margin		-0.006		-0.006		-0.006
		(0.008)		(0.008)		(0.008)
Negative Income		1.029***		1.029***		1.029***
		(0.052)		(0.052)		(0.052)
Leverage 1		4.007***		4.007***		4.007***
		(0.199)		(0.199)		(0.199)
Log(Tot. Assets)		0.129***		0.129***		0.129***
		(0.019)		(0.019)		(0.019)
Year FE	✓	✓	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓	✓	✓
Country FE	✓	✓	✓	✓	✓	✓
N	41965	25546	41965	25546	41965	25546
R^2	0.066	0.182	0.066	0.182	0.066	0.182

Table A7: **Zombie Firms, Distressed Firms, and Recession Events.** This table presents logistic regressions. The dependent variable is a binary variable that is equal to 1 if a company is a zombie (Panel A) or distressed (Panel B) and 0 otherwise. A company is classified as zombie following R. Banerjee and Hofmann (2020) and distressed following Altman (1968). Dot-com Bubble, (2000-2001), Financial Crisis, (2008-2009), and Debt Crisis, (2011-2012) are dummy variables for the respective recession events plus three years after the event. Leverage 1 is the sum of long-term debt and debt in current liabilities divided by total assets. Standard errors in parentheses are clustered at firm-level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

A.3 Identifying Zombies

The strand of literature focusing on zombie companies provides different approaches to identifying a company as a zombie, each one of them with its own advantages and drawbacks. The zombie definition itself, together with data limitations, explains the existing measurement challenges.

One of the first measures originates from the study of Caballero, Hoshi, and Anil K. Kashyap (2008). The authors classify a company as a zombie whenever it receives subsidized credit at an interest rate that is below the one applied to the most creditworthy companies. The actual interest payments made by the companies are then compared to an estimated benchmark, R^*, based on the firm's debt structure and market prime rate. The minimum required interest payment for each firm i in year t, R^*_{it}, is:

$$R^*_{it} = rs_{t-1}BS_{it-1} + \Big(\frac{1}{5}\sum_{j=1}^{5} rl_{t-j}\Big)BL_{it-1} + rcb_{\text{min last 5 yr, t}} \times Bonds_{it-1}. \tag{1}$$

where BS_{it}, BL_{it}, and $Bonds_{it}$ represent short-term loans (less than one year), long-term bank loans (more than one year), and total bonds outstanding (including convertible bonds and warrant-attached bonds), respectively, for firm i at end of year t; while rs_t, rl_t, and $rcb_{\text{min last 5 years, t}}$ represent the average short-term prime rate in year t, the average long-term prime rate in year t, and the minimum coupon rate on any convertible corporate bond issued in the last five years before t.[14]

In this study, given that we examine zombie companies from a sample of publicly listed firms in Europe we do not replicate the above measure. As noted by R. Banerjee and Hofmann (2018), by employing this measure one would encounter three potential limitations: *1)* identifying with precision the subsidized credit granted by the banks to

[14]From Caballero, Hoshi, and Anil K. Kashyap (2008) we recall that the authors do not know the exact interest rates on specific loans, bonds, or commercial paper, nor the exact maturities of any of these obligations.

the companies would be a challenge, *2)* banks may grant subsidized credit for other reasons, and *3)* when interest rates are very low for longer periods, subsidized lending rates would have to be near zero or even negative.

The recent literature adopts a measure that relies on the accounting information of the firm, its age, and in its future growth potential. The most widely used measure evolve around the interest coverage ratio, initially used by McGowan, Andrews, and Millot (2017) and central banks' reports.[15] A company, in this case, is considered a zombie whenever its ICR_{it} is below 1 for 3 consecutive years and its age is at least 10 years. In R. Banerjee and Hofmann (2018) and R. Banerjee and Hofmann (2020), the latter measure is complemented with the company's future growth potential, captured with the Tobin's q, to account for investors' sentiment. In Acharya, Crosignani, et al. (2020) a company is considered a zombie if it meets two criteria: *1)* the firm's ICR is below median and its leverage ratio is above median, *2)* the share of interest expenses relative to the sum of its outstanding loans, credit, and bonds in a given year is below the interest paid by the most creditworthy firms (Acharya, Eisert, et al. 2019).

[15] Bank of England 2013 report "Inflation Report" and Bank of Korea 2013 report "Financial Stability Report". The reports are available from: `https://www.bok.or.kr/eng/bbs/E0000737/view.do?nttId=194333&menuNo=400219` and `https://www.bankofengland.co.uk/-/media/boe/files/inflation-report/2013/august-2013.pdf`.

References

Acharya, Viral V., Matteo Crosignani, et al. (2020). “Zombie Credit and (Dis-)Inflation: Evidence from Europe”. In: *National Bureau of Economic Research* 27158.

Acharya, Viral V., Tim Eisert, et al. (2019). “Whatever it Takes: The Real Effects of Unconventional Monetary Policy”. In: *The Reveiw of Financial Studies* 32.9, pp. 3366–3411.

Aghion, Philippe, Emmanuel Farhi, and Enisse Kharroubi (2019). “Monetary Policy, Product Market Competition and Growth”. In: *Economica* 86.343, pp. 431–470.

Altavilla, Carlo, Lorenzo Burlon, et al. (2021). “Is there a zero lower bound? The effects of negative policy rates on banks and firms”. In: *Journal of Financial Economics, forthcoming.*

Altavilla, Carlo, Giacomo Carboni, and Roberto Motto (2015). “Asset purchase programmes and financial markets: lessons from the euro area”. In: *ECB Working Paper Series* 1864, pp. 1–52.

Altman, Edward I. (1968). “Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy”. In: *The Journal of Finance* 23.4, pp. 589–609.

Andrews, Dan and Filippos Petroulakis (2017). “Breaking the Shackles: Zombie Firms, Weak Banks and Depressed Restructuring in Europe”. In: *OECD Working Papers* 1433.

Asquith, Paul, Robert Gertner, and David Scharfstein (1994). “Anatomy of Financial Distress: An Examination of Junk-Bond Issuers”. In: *The Quarterly Journal of Economics* 109.3, pp. 625–658.

Banerjee, Abhijit V. and Esther Duflo (2005). “Growth Theory through the Lens of Development Economics”. In: *Handbook of Economic Growth.* Ed. by Philippe Aghion and Durlauf Steven. Vol. 1A. Elsevier, pp. 473–552.

Banerjee, Ryan and Boris Hofmann (2018). “The rise of zombie firms: causes and consequences”. In: *BIS Quarterly Review* September.

— (2020). “Corporate zombies : Anatomy and life cycle”. In: *BIS Working Papers* 882.

Bauer, Michael D and Glenn D Rudebusch (2013). “The Signaling Channel for Federal Reserve Bond Purchases”. In: *Federal Reserve Bank of San Francisco*. Federal Reserve of San Francisco April.

Bernanke, Ben and Alan Blinder (1988). “Credit, Money, and Aggregate Demand”. In: *American Economic Association* 78.2, pp. 435–439.

Bhamra, Harjoat S., Christian Dorion, et al. (2018). “Low Inflation: High Default Risk and High Equity Valuations”. In: *National Bureau of Economic Research* 25317.

Bhamra, Harjoat S., Adlai J. Fisher, and Lars Alexander Kuehn (2011). “Monetary policy and corporate default”. In: *Journal of Monetary Economics* 58.5, pp. 480–494.

Borio, Claudio (2018). *A blind spot in today’s macroeconomics?* Tech. rep. January. Bank for International Settlements, pp. 1–14.

Caballero, Ricardo J., Takeo Hoshi, and Anil K. Kashyap (2008). “Zombie lending and depressed restructuring in Japan”. In: *American Economic Review* 98.5, pp. 1943–1977.

Campbell, John Y., Jens Hilscher, and Jan Szilagyi (2008). “In search of distress risk”. In: *Journal of Finance* 63.6, pp. 2899–2939.

Cleary, Sean (1999). “The relationship between firm investment and financial status”. In: *Journal of Finance* 54.2, pp. 673–692.

De Martiis, Angela, Thomas Heil, and Franziska Peter (2020). “Are you a Zombie? A Supervised Learning Method to Classify Unviable Firms and Identify the Determinants”. In: *SSRN Electronic Journal* January.

De Santis, Roberto A. et al. (2018). “The impact of the corporate sector purchase programme on corporate bond markets and the financing of euro area non-financial corporations”. In: *ECB Economic Bulletin* 3.4, pp. 66–84.

Duval, Romain, Gee Hee Hong, and Yannick Timmer (2020). “Financial Frictions and the Great Productivity Slowdown”. In: *Review of Financial Studies* 33.2, pp. 475–503.

European Central Bank (2016). “The corporate bond market and the ECB’s corporate sector purchase programme”. In: *ECB Economic Bulletin* 5 Box 2, pp. 3–6.

Fairfield, Patricia M. and Teri Lombardi Yohn (2001). “Using asset turnover and profit margin to forecast changes in profitability”. In: *Review of Accounting Studies* 6.4, pp. 371–385.

Fazzari, Steven, R. Glenn Hubbard, and Bruce Petersen (1988). “Investment, Financing Decisions, and Tax Policy”. In: *American Economic Review* 78.2, pp. 200–205.

Gern, Klaus Jürgen et al. (2015). “Quantitative Easing in the Euro Area: Transmission Channels and Risks”. In: *Intereconomics* 50.4, pp. 206–212.

Giannetti, Mariassunta and Andrei Simonov (2013). “On the real effects of bank bailouts: Micro-evidence from Japan”. In: *American Economic Journal: Macroeconomics* 5.1, pp. 135–167.

Gordon, M. J. (1971). “Towards a Theory of Financial Distress”. In: *The Journal of Finance* 26.2, pp. 347–356.

Grosse-Rueschkamp, Benjamin, Sascha Steffen, and Daniel Streitz (2017). “Cutting Out the Middleman: The ECB as Corporate Bond Investor”. In: *Available at SSRN* October.

Hoshi, Takeo (2006). “Economics of the living dead”. In: *Japanese Economic Review* 57.1, pp. 30–49.

Igan, Deniz, Gee Hee Hong, and Do Q Lee (2021). “Zombies on the Brink: Evidence from Japan on the Reversal of Monetary Policy Effectiveness”. In: *IMF Working Papers* 044.

Jordà, Òscar et al. (2020). “Zombies at Large? Corporate Debt Overhang and the Macroeconomy”. In: *National Bureau of Economic Research* 28197.

Joyce, Michael et al. (2012). “Quantitative easing and unconventional monetary policy - an introduction”. In: *The Economic Journal* 122.564, F271–F288.

Kane, Edward J (1989). “The High Cost of Incompletely Funding the FSLIC Shortage of Explicit Capital”. In: *Journal of Economic Perspectives* 3.4, pp. 31–47.

Kaplan, Steven N. and Luigi Zingales (1997). “Do investment-cash flow sensitivities provide useful measures of financing constraints?” In: *Quarterly Journal of Economics* 112.1, pp. 169–213.

Kashyap, Anil K and Jeremy C Stein (1994). “The impact of monetary policy on bank balance sheets”. In: *National Bureau of Economic Research* 4821.

Krishnamurthy, Arvind, Stefan Nagel, and Annette Vissing-Jorgensen (2018). “ECB Policies Involving Government Bond Purchases: Impact and Channels”. In: *Review of Finance* 22.1, pp. 1–44.

Krishnamurthy, Arvind and Annette Vissing-Jorgensen (2011). “The Effects of quantitative easing on interest Rates: Channels and implications for policy”. In: *National Bureau of Economic Research* 17555, pp. 215–287.

Laeven, Luc, Glenn Schepens, and Isabel Schnabel (2020). “Zombification in Europe in times of pandemic”. In: *VoxEU.org* 11 October.

Lenza, Michele, Huw Pill, and Reichlin Lucrezia (2010). “Monetary policy in exceptional times”. In: *Economic Policy* 25.62, pp. 295–339.

McGowan, Müge Adalet, Dan Andrews, and Valentine Millot (2017). “Insolvency Regimes, Zombie Firms and Capital Reallocation”. In: *OECD Working Papers* 1399.

McGowan, Müge Adalet, Dan Andrews, and Valentine Millot (2018). "The walking dead? Zombie firms and productivity performance in OECD countries". In: *Economic Policy* 33.96, pp. 685–736.

Midrigan, Virgiliu and Daniel Yi Xu (2014). "Finance and Misallocation: Evidence from Plant-level Data". In: *American Economic Review* 104.2, pp. 422–458.

Obstfeld, Maurice and Romain Duval (2018). "Tight monetary policy is not the answer to weak productivity growth". In: *VoxEU.org* April.

OECD (2021a). *Composite leading indicator (CLI)*. DOI: `10.1787/4a174487-en`.

— (2021b). *Long-term interest rates (indicator)*. DOI: `10.1787/662d712c-en`.

— (2021c). *Short-term interest rates (indicator)*. DOI: `10.1787/2cc37d77-en`.

Peek, J and Eric Rosengren (1995). "Bank Lending and the Transmission of Monetary Policy". In: *Is Bank Lending Important for the Transmission of Monetary Policy?* Ed. by J. Peek and E. S. Rosengreen, pp. 47–68.

Rodnyansky, Alexander and Olivier M. Darmouni (2017). "The effects of quantitative easing on bank lending behavior". In: *Review of Financial Studies* 30.11, pp. 3858–3887.

Satu, Nurmi, Juuso Vanhala, and Matti Verín (2020). "The life and death of zombies - evidence from government subsidies to firms". In: *Bank of Finland Discussion Papers* 8.

Schivardi, Fabiano, Enrico Sette, and Guido Tabellini (2021). "Credit Misallocation During the European Financial Crisis". In: *The Economic Journal* ueab039.

Valta, Philip (2016). "Strategic Default, Debt Structure, and Stock Returns". In: *Journal of Financial and Quantitative Analysis* 51.1, pp. 197–229.

Vayanos, Dimitri and Jean-Luc Vila (2009). "A Preferred-Habitat Model of the Term Structure of Interest Rates". In: *National Bureau of Economic Research* 15487.

Zmijewski, Mark (1984). "Methodological Issues Related to the Estimation of Financial Distress Prediction Models". In: *Journal of Accounting Research* 22, pp. 59–82.